

Universität Bremen

Faculty 3 - Mathematics and Computer Science



Human-Computer Interaction
University Bremen

Master Thesis

Sociability of Mobile Listening Technologies in Public Urban Surroundings

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December 29, 2021

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University Bremen, December 2021

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Acknowledgement

I would like to thank all the people that have aided and supported me during the creation of my magnum opus.

First and foremost, I want to thank my parents for their absolute belief in me. For their support and love which I could feel all the way.

I want to thank all my friends and flatmates that have cared for me through good and bad times. That had given me space when I needed it and always wholeheartedly embraced me again.

I want to thank Tim for always turning my frustration into laughter and joy. For reminding me of the good in the bad. For being the best friend, one could wish for.

I want to thank Luca for filling me with love and compassion. For never failing to cheer me up or calm me down when I needed it. For always giving me a reason to cherish life and look forward to days to come.

I want to thank all participants of my user study for their time, their cooperation and their advice.

I want to thank Johannes Schöning for giving me the freedom to develop my ideas, and never failing to give me a push in the right direction when I struggled to see my path clearly.

Big thanks go out to Silke Völkers and Nadja DaSilva from the faculty 3 examination office. Throughout my studies they have shown mad commitment, fairness, helpfulness and kindness towards all students while being highly professional and competent. They have turned all administration work into delight and I was always looking forward towards my next appointment.

I want to thank Earl Grey tea for the hundreds of liters I enjoyed while writing this thesis.

Pitt Michelmann
Bremen, December 29, 2021

Abstract

This thesis analyzes the social aspects of modern mobile music consumption in public spaces. We researched the changes in social behavior brought by advancing technologies to identify social challenges and limitations of modern consumption patterns. We found factors in modern mobile music consumption limiting social interactions and exchange and serendipitous music discovery.

This thesis aims to devise universal best practices and design paradigms for creating applications focused on improving mobile listening experiences and engaging the isolating effects of mobile listening practices. We analyzed studies on location-based applications, music discovery and urban computing and collected their design choices and findings. As an exemplary implementation, we derived our own concept and prototype application, Soundtrack Of Our Lives (SOOL). SOOL enables the real-time sharing of music playback with collocated strangers in urban places using mobile phones and Spotify. We conducted a user study using our prototype and discuss its potential to overcome public social isolation and instead create a sense of social cohesion. This study's findings should help to create guidelines for future research in the field of location-based mobile music sharing in urban public environments.

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Chapter 1

Introduction

With the commercial success of the Sony Walkman since its release in 1979, music listening practices have undergone fundamental changes. Before, music consumption was mainly bound to specific locations like bars, clubs, cars or home stereos (Crossen et al., 2006). Now people carried and chose their own soundtrack on the go. Since then, mobile music listening technology has significantly advanced. The miniaturization of devices and the development of storage formats like MP3 allowed people to carry their entire music collections with them at all times. The integration of music players into internet-enabled mobile devices and the growing availability of online music streaming services further extended the portability of music, giving users access to tens of millions of songs. Mobile music consumption is now a ubiquitous aspect of most peoples lives (Åman et al., 2015). In urban surroundings, mobile music listening offers people a tool to 'tune out' and distance themselves from their surroundings (Watson et al., 2016). Through the consumption of music, they gain the means to alter or control their moods, to distract themselves and to manage their interactions with other city dwellers by creating their own private space with its own underlying soundtrack (Bull, 2006; Bull, 2007; Bull, 2010).

Research suggests that the growing popularity of smartphones combined with advancing mobile adaptation of social networks leads to self-isolating social behaviors. Instead of concerning themselves with their surrounding, people use their mobile phones to connect to their distant, already existing social circle. This act of 'cocooning' inhibits mutual awareness and social participation in urban life (Bassoli et al., 2007; Crawford, 2008; Seeburger et al., 2010; Kirk et al., 2016; Paasovaara et al., 2018). Alongside modern music recommendation algorithms, this not only keeps people in their own social and auditory bubble but also inside of a musical filter bubble (Taramigkou et al., 2013). These algorithms analyze the user's musical preferences, aiming to generate recommendations matching the user's taste. "Hence, most online recommender systems do not enable users to experience novelty and serendipity in the provided suggestions" (Taramigkou et al., 2013).

While these modern advancements in the field of mobile music listening technology seem to have a mainly disruptive influence on social interactions and integration, there has been growing

research interest in how the widespread use of location-aware and internet-enabled mobile devices could provide opportunities to encourage a sense of commonality and civic participation in public spaces (Hakansson et al., 2005; Watson et al., 2016; Seeburger, 2012; Kirk et al., 2016). So-called People-Nearby Applications (PNAs) aim to form connections between collocated strangers using the user's location.

In several PNA studies, users expressed fear of sharing details of their identity with nearby strangers but are in general comfortable with broadcasting anonymous information like their taste in music (Bassoli et al., 2006; Seeburger et al., 2010; Kirk et al., 2016). DeNora (2006) highlights the social aspects of music consumption and the high potential of music sharing for social bonding. Some existing research on PNAs explores this potential to facilitate meaningful social connections in crowded public places. Employing different tactics and technologies, researchers showed the aptitude of mobile music sharing as a social icebreaker to connect friends and strangers alike and create a sense of commonality through the social capabilities of music. Some of these tactics include sharing song metadata with collocated strangers (Seeburger et al., 2012; Baumann et al., 2007), some exchange files (Jacobsson et al., 2005; Bassoli et al., 2006) while others use specific locations to 'drop' and exchange songs (Åman et al., 2015). Evaluating existing studies, we see great potential in location-based mobile music sharing and the need for further research in this field.

We collected common approaches and design choices made the examined studies to formulate best practices and design paradigms when creating PNAs in the field of music discovery. We distilled our findings to create our own concept for a location-based mobile music discovery application called Soundtrack Of Our Lives (SOOL). SOOL is a mobile phone application that enables users to broadcast their Spotify playback in a certain range so other application users can tap into their music to co-listen. In contrast to other existing applications, SOOL uses state of the art technology and is the first application to adapt the modern practice of music streaming over the internet. It also enables synchronized playback, so far only seen in the tunA (Bassoli et al., 2006) study.

In the following chapters, we will further shine a light on the social aspects of music consumption and the challenges of modern music consumption patterns. We will then review existing research approaches and their findings. Subsequently, we will introduce SOOL, its features and tactics to engage found challenges. We tested our concept by creating a prototype that implements some of its features. We conducted a qualitative user study to verify our design choices and the connecting qualities of our approach. Finally, we will discuss our findings and their significance for future work in the field of designing location-based mobile music listening applications with the aim of improving the sociality of modern listening habits.

Chapter 2

Background

In this chapter, we aim to inform about the social significance of music and the influence of changing consumption technologies on social practices surrounding music consumption. We will discuss the challenges and opportunities of modern music listening practices and present a selection of studies addressing these challenges with the aim of improving mobile music listening experiences and social interaction in public environments.

2.1 Sociality of Music

Music plays a huge role in our private and social lives. According to the *Music Listening 2019* report (IFPI, 2019), people on average spend 18 hours a week listening to music. People not only see music as a tool for their own entertainment or as means for mood control and alteration but also as an expression of their character (DeNora, 2006; Voida et al., 2006). One's own taste in music is an expression of one's personality and self-identity (Hargreaves et al., 1999). Music makes people socially interact with each other through showing, sharing and talking about music. People - even strangers - can instantaneously bond over sharing the same taste in music. People also take special pride in getting positive comments on their choice of music (Brown et al., 2006).

2.1.1 Social Practices Around Changing Technologies

The sociability of music consumption and the surrounding social practices heavily depend on the established and available music consumption technologies. As new technologies emerge, old practices adapt, change, or new practices develop (see figures 2.1 and 2.2) (Leong et al., 2013). In the following, we will give a short historical overview of music technology (and its accompanying social importance.

Live Performances

Before the invention of recording technologies, music had to be performed live by artists. For people lacking the personal skill, acquaintances or musical instruments, music could mainly be enjoyed at public locations, venues or social gatherings like in church, at concerts and at public festivities (Coleman, 2005, p. 1).

Phonograph

The phonograph, developed by Thomas Edison in 1877, was the first reliable music recording and playback technology. This technology was expensive, of poor quality and limited in terms of capacity until the development of disc-based recording mediums in the beginning of the 20th century (Coleman, 2005, pp. 2, 10 ff.). Subsequent enhancements of technology and the transition from glass and rubber-based materials to shellac and vinyl in 1949 allowed for longer playback duration, which in turn promoted the creation and recording of whole albums on so-called Long Plays (LPs). This established disc records as the first widespread commercial private music listening technology and, for the first time, enabled the mass replication and distribution of music records. Vinyl records were dominant in the music industry market until the late 1970s (Coleman, 2005, pp. 34 ff., 52 ff.).

Public Radio

The first clear transmission of human speech via radio waves was made in 1919. Following the development of the first recording technologies, public radio broadcasts gained popularity. Broadcasts reached millions of people weekly and allowed for a wholly new stage for music discovery and distribution. Additionally, besides the cost of purchase, listening to music was free of charge. Since the 1930s, the stationary radio receiver had become a standard domestic technology. In the 1940s portable radios emerged as the first widely used mobile music listening technology, reaching a status as a standard household appliance in the 1960s. As radios run a predetermined schedule and private recording technologies were mostly unavailable, people would sometimes gather at specific times to listen to programs or listen by themselves and later discuss the programs (Coleman, 2005, pp. 29, 86 ff.).

In contrast to radio programs, physical copies of music on vinyl and later cassettes and Compact Discs (CDs) allowed for consumers to assemble their own private music collections, which could be listened to at any time. Sharing music with other people was now no longer bound to a specific place or time. People would discover new music by showing and borrowing their collections from each other (Voida et al., 2006).

Cassette Tapes

The invention of cassette tapes in 1963 was the first recording technology to support easy and reliable homemade music copying. It quickly reached a dominant status as a recording medium, lasting until the 1990s (Coleman, 2005, pp. 155 ff.). With the gaining popularity of cassettes and the development of high-quality portable cassette players, people started to adopt new habits of consuming music. Consumers could now record music they liked from the radio, copy other people's music or create compilations from their own collections. People started to create compilations for particular occasions, locations, social circles or persons. These self-created 'mixtapes' not only serve the purpose of enjoying and bonding over the shared music but also as a portrayal of self by revealing part of one's musical identity (Voida et al., 2006).

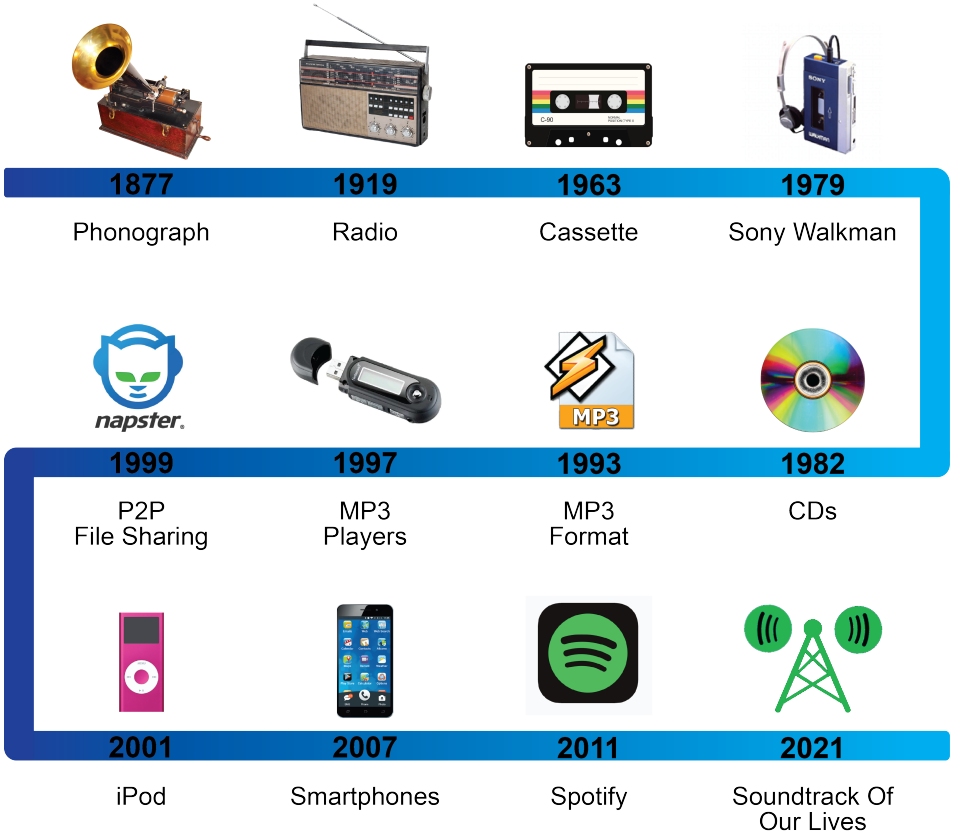


Figure 2.1 Timeline of Mobile Music Listening

Sony Walkman

The Sony Walkman, released in 1979, marked the first widely available and successful private mobile listening technology. Consumers could now listen to their favorite music anywhere they

went. People wearing headphones in public spaces became a common sight (Coleman, 2005, p. 158).

> *"In the past hundred years, the predominant situation of listening to music has moved successively from one of the public, shared consumption of live production, to one of the potentially shared consumption of recorded or broadcast production, to private consumption"*
>
> (Stewart et al., 2018)

From a technological point of view, the step from portable radio-cassette-players and car stereos to the Walkman was a small, rather regressive one. In contrast, its influence on listening practices was extremely high (Hosokawa, 1984).

CDs

The Compact Disc, developed in 1982, was the first medium to enable the saving of digital music. It also was the first medium to enable easy and accurate song skipping - elevating the importance of single tracks in relation to the whole album or collection. As CD players have no in-built function to record or copy their own music (like cassette tape players often had), consumers had to come up with their own solutions for deploying this technology. CD 'burning' using a computer had a huge impact on future distribution and music sharing as it enabled digital copying and rearranging of songs, which is far more reliable and lossless than copying cassettes (Coleman, 2005, pp. 164 ff.). The CD quickly replaced vinyl records and cassettes as standard recording technology.

> *"After 1991, vinyl virtually disappeared"*
>
> (Coleman, 2005, p. XV)

MP3

The development and proliferation of digital formats in the 1990s greatly increased music storage capabilities and availability. The establishment of the MPEG (Motion Picture Experts Group) Layer-3 (MP3) format marked the first non-physical music saving medium and allowed for music to be effortlessly transported, copied and shared and is used to this day. MP3 players allow for mobile playback of MP3 files. The files are typically copied from computers where they can be nearly instantaneously accessed or sent over great distances using the internet. Before, the transportability of music was limited by the size of storage mediums (e.g. how many CDs you could carry around). Now it is only limited by the growing storage capabilities of disc drives or solid-state drives, used in MP3 players (Coleman, 2005, pp. 184 ff.). In 2001 Apple released the iPod, which was able to store thousands of songs, effectively enabling consumers to carry their whole music collection around wherever they went (Voida et al., 2006; Leong et al., 2013).

Peer-To-Peer Downloading

In 1999 the peer-to-peer downloading platform Napster gained popularity among college students. It allowed the sharing of MP3 files over the internet for free. Users could search for and download tracks and even whole albums. It also allowed for the sharing of rare versions of songs like live recordings, alternate cuts or demos, which was highly celebrated among music enthusiasts (Coleman, 2005, pp. 189 ff.).

It was shut down in 2001 due to copyright violations but at that time counted up to 24.1 million active users (*Napster* 2021).

> *"Napster and other file-sharing schemes aren't just about stealing music; file sharing encourages listeners to pursue hunches, check out tips, indulge whims, and develop new enthusiasms"*
>
> (Coleman, 2005, p. XVi)

iTunes Store

In 2003 Apple started the iTunes Store, a digital music library that offers the possibility to buy and download songs. Consumers would pay 99 cents for a song and could then download the song to any iPod player, to up to three apple computers and burn 10 CDs with the same playlist (Coleman, 2005, pp. 205 ff.). The development of internet infrastructure and the rising demand for mobile internet in the mid-2000s lead to great advancements in bandwidth speeds, internet coverage and ubiquitous availability of mobile internet. This, in turn, paved the way for the thriving of mobile music streaming services.

> *"I think the CD system will be around for another 20 years. There is no reason for it to go away. Certainly not because of the internet - which will never represent mass tonnage market for albums. It will only be good for singles or free music"*
>
> - Russ Solomon, Tower Records, 2001

Modern Music Streaming Services

In 2005 Pandora established itself as a new subscription-based music streaming service in the United States, Australia and New Zealand. Its main focus is on music discovery using the 'Music Genome Project' - software that analyzes and interconnects songs and genres by musical traits - and presents the user with the possibility to create 'Stations' based on their musical taste and listening history. Its user count peaked in 2016 with 81 million users, but that count has been declining ever since as other streaming services with vaster music libraries like Spotify and Apple Music became increasingly popular (see figure 2.3) (*Pandora (streaming service)* 2021).

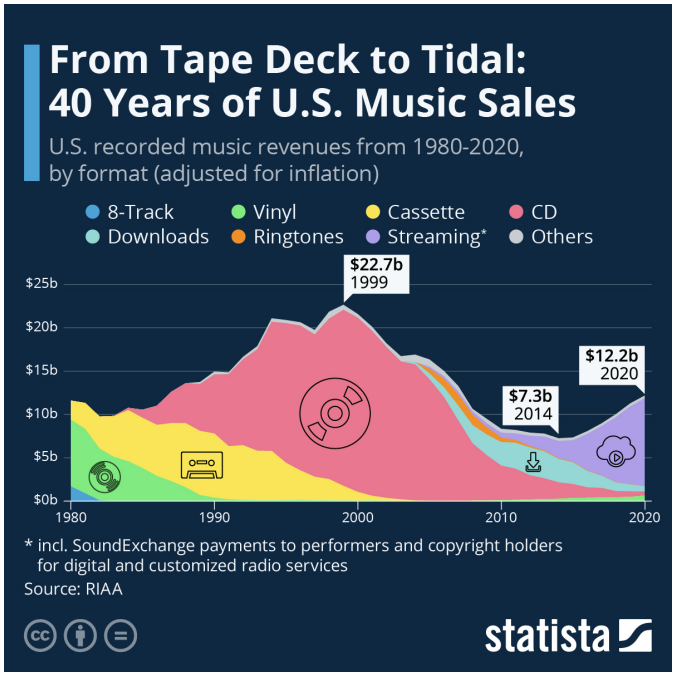


Figure 2.2 Music revenues by format (*Infographic* 2021)

For consumers, music streaming services provide ubiquitous access to all of their music across a variety of technological platforms that can access the internet. Music is no longer owned and people's personal music libraries are now simply a part of bigger, global repositories. Personal taste and musical identity are no longer defined by the music you own or have access to. The music is accessible to anyone, the uniqueness of one's own taste in music lies in the selections you make. Music collections are now replaced by deliberately crafted playlists. Collocation is no longer needed to recommend and share music. Sharing has never been as easy as it is with streaming. Additionally, some impression management practices have now shifted to social media, broadcasting one's taste and music identity publicly (Leong et al., 2013).

2.1.2 Private Mobile Music Listening

Bull (2006) stated that individual mobile music listening in crowded public places is creating a 'private auditory bubble' used to set up boundaries and negotiate one's own space by creating their own underlying soundtrack. He further described this as a behavior of self-prioritization, giving people the means to manage and control their private spaces (Bull, 2010). D'Arcangelo (2005) remarked: "People 'sing their unique birdsong' through their personal music collections, differentiating themselves from the crowd and mapping the unique topography of their personal musical terrain".

Since the release of the first smartphones in 2007, MP3 players have slowly been replaced by the music playing capabilities of multi-purpose mobile phones. In 2010, the sales of the iPhone surpassed those of the iPod, mirroring the rising integration of music player functionality in mobile phones. In more recent years, the proliferation of internet-enabled mobile phones, increasingly fast and reliable mobile internet connections and the vast repositories of music streaming services allow for high quality continuous mobile music streaming. In public, people now would listen to music, engage in social networks or communicate with friends, all using one device. Social interaction has shifted from collocated to distant. Crawford (2008) calls this practice 'cocooning'. When in crowded public areas, people rather connect to their existing social circles using their smartphones. Combined with the auditory bubble created by personal mobile music listening, people drown out their surroundings, limiting social participation and possibilities for social interaction. Seeburger et al. (2012) argue that people are still naturally curious about other collocated strangers. They suggest that by building upon existing consumption practices, the possibilities of modern technology can be used to create a sense of commonality between strangers while enabling people to still inhabit their cocoons. "Portable music can be used to connect - or simply associate - with other people as a common ground for interaction" (Seeburger et al., 2010).

2.1.3 Spotify

In 2020 62% of the global recorded music revenues were created by music streaming (*IFPI GLOBAL MUSIC REPORT* 2021). With 329 million users and 144 million paid subscribers, Spotify is the most popular music streaming service worldwide (see figure 2.3) (Gomes et al., 2021). At first, Spotify was set on providing legal and free means of streaming music in a so-called 'freemium' concept. Users could use the service for free but had to pay for premium features like better sound quality, download functionality and disabling advertisement. Nowadays, it is mostly subscription oriented with only 13% of its income being generated through ads (Fleischer, 2021).

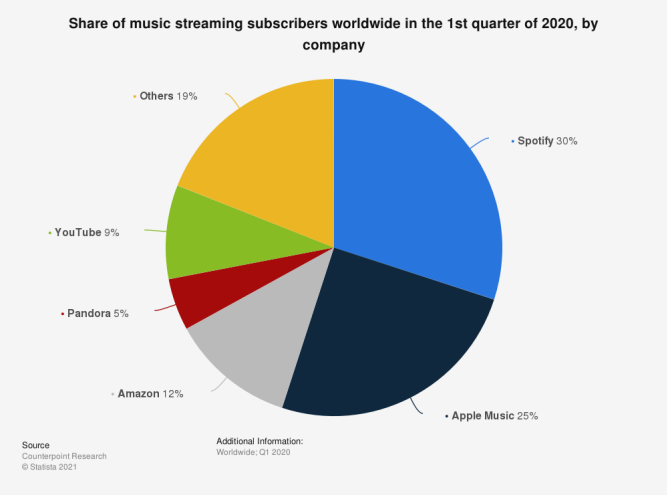


Figure 2.3 Share of music streaming subscribers worldwide in the 1st quarter of 2020, by company (*Music streaming market share* 2021)

2.1.4 Musical Filter Bubbles

In the last years, Human-Computer Interaction (HCI) research in the field of music consumption was mostly preoccupied with improving profile management and music recommender systems (Kirk et al., 2016). Recommender systems are used to deal with the enormous amounts of information available on the internet and to search for information personally suitable for a specific user. They gather information about the user's online behavior like search history, cookies, view and listening events, user feedback or expressed taste to generate a model of the user's preferences. This model is then used to suggest content that aligns with the user's personal interests like in search engine results, news articles, online shopping, movies and music (Melville et al., 2010). While this automatic filtering is necessary to at all be able to traverse the vast datasets provided by the internet, 'users are constantly missing something without noticing it, and, on the other, they are getting continuously the same type of recommendations' (Maccatrozzo, 2012). Pariser (2011) called this phenomenon the 'filter bubble' - an over-personalization effect that hides content that does not completely fit a user's profile, therefore limiting the chance of serendipitous encounters.

Recent research shows that recommender systems of online music bibliographies like Spotify lead to less diverse consumption patterns and that, in turn, users who have a very diverse taste in music fall back to manually picking music choices (Anderson et al., 2020). Many music listeners wish to explore new music to "expand their interests, acquire a more divergent view of music and identify novel and serendipitous content" (Taramigkou et al., 2013). They often aim to extend their personal view and explore what other users like to empathically connect to and understand

them (Taramigkou et al., 2013). There have been algorithmic attempts to counter the isolating effects of recommender systems with moderate success (Flexer et al., 2008; Zhang et al., 2012; Taramigkou et al., 2013). Other promising research proposes the use of location-based mobile media applications to produce serendipity, positive chance encounters, using the musical choices of collocated strangers (Jacobsson et al., 2005; Bassoli et al., 2006; Seeburger, 2012; Åman et al., 2015; Kirk et al., 2016). They devised applications that enable sharing of music with random collocated people. Here the presented music is being filtered by location, which is a criterion totally unrelated to the music's characteristics and is therefore a promising tool to burst the bubble.

2.2 Related Work

There are some studies introducing location-based applications that aim to connect collocated strangers through the use of music sharing. These systems either allow synchronous music sharing, sharing of files or recommendations and messaging with either friends or strangers. In the following, we like to present some of these studies and prototypes that inspired the creation of SOOL.

tunA Bassoli et al. (2006) uses handheld WiFi-enabled PDAs to enable real-time synchronous playback with nearby strangers (see figure 2.4). Users can listen to MP3 encoded files they have previously transferred onto their devices and can broadcast their playback. When within range, other devices can download the MP3 files, and synchronous playback can be started. An inbuilt messenger allows for further personal exchange. In a user study analyzing the socializing effects of the application, the subjects "showed how sharing music constitutes a form of social bonding with strangers and a fun experience with friends" (Bassoli et al., 2006).

As a successor to tunA, Baumann et al. (2007) created **BlueTunA**, an application for mobile phones that uses Bluetooth to exchange music metadata with nearby strangers (see figure 2.5). The application creates an automated list of either MP3 files stored on the device or of the user's Last.FM profile. It then compares to the lists of other BlueTunA users in the vicinity, displaying users who share a common taste in music. Users can then review other users' songs metadata and decide to start messaging over Bluetooth. The application has not yet been evaluated.



Figure 2.4 tunA prototype by Bassoli et al. (2006)



Figure 2.5 BluetunA prototype by Baumann et al. (2007)

Pocketsong is a mobile phone app developed by Kirk et al. (2016) as a technological probe to understand the real-world user experience of mobile music sharing in social and public settings (see figure 2.6). Pocketsong enables the asynchronous sharing of songs at specific locations. Users can 'drop' songs or gift songs to specific users, which can then be encountered or collected when other users approach that location. Due to copyright limitations, only short song introductions of at most 15 seconds could be shared. In a user study Kirk et al. (2016) analyzed some social aspects of location-based anonymous music sharing applications. They determined that the shared songs do not necessarily reflect their real taste but rather are a form of technology-mediated self-expression management. Additionally, users expressed the wish to further contextualize their choices as well as create a richer dialogue.



Figure 2.6 Pocketsong Snoop Interface (Kirk et al., 2016)

OUTMedia (Åman et al., 2015) is an application that uses Augmented Reality (AR) to link music to specific places (see figure 2.7). In concept, the application should allow for users to freely create AR objects with songs and annotations appended for other users to discover. In their study, however, participants could only interact with content already created by the researchers. Åman et al. (2015) analyzed how users experienced the service in terms of serendipity, overall user experience and engagement. They found that the application supported interaction and discovery well. Participants maintained sustained attention, enjoying the free discovery of music. They also attributed the system a high level of usability. They concluded "that music can not only take people somewhere else (the created virtual place or 'non-place') but also enriches the experience of places where it is attached" (Åman et al., 2015).



Figure 2.7 OUTMedia Interface: AR view (left) and player view (right) (Åman et al., 2015)

Capital Music (Seeburger et al., 2012) is a mobile phone application that allows strangers, collocated at specific locations, to share metadata of songs they currently listen to in real-time (see figure 2.8). They conducted a Wizard of Oz (WoZ) user study to analyze the user experience

of the app in terms of usability, the users' curiosity about collocated strangers as well as the felt engagement of technology-mediated social interactions. Their findings revealed that the app stimulates curiosity and makes the users more conscious about their surroundings as well as collocated people. Participants saw a potential use of the app for music recommendation and discovery as well as a suitable icebreaker for sparking up conversations with strangers in public environments. They saw the high potential of location-based music sharing applications to create a sense of commonality while still allowing people to 'cocoon' themselves in public. Similar to the findings of Kirk et al. (2016), Seeburger et al. (2012) determined that users listening behavior changes when their choices are being publicly broadcasted. People curate their playlists more carefully, choosing either songs that they feel represent themselves best or songs that match already displayed musical tastes.

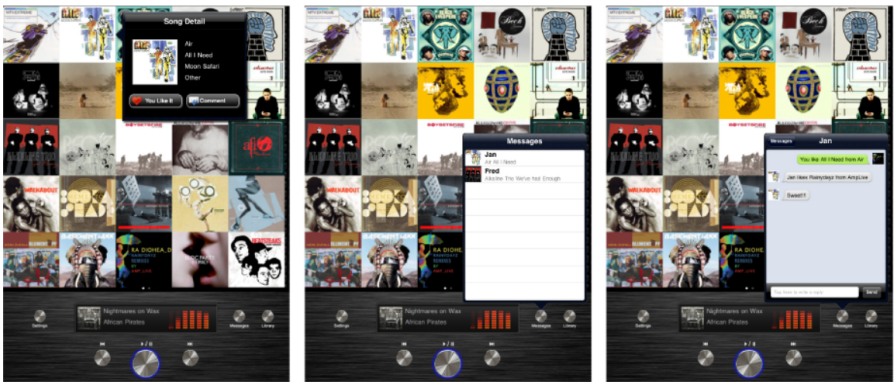


Figure 2.8 Capital Music Interface (Seeburger et al., 2012)

Stewart et al. (2018) studied the music listening habits of young people in rural areas and the potential of modern music listening technologies to connect people. They created **Colisten**, a mobile phone application that provides the possibility for location-independent synchronous playback between friends using the Spotify music streaming service (see figure 2.9). They conducted a user study with 26 participants from rural areas aged 9 to 15 years, where the participants were exclusively pairs of friends. The study reports that most participants were interested in listening to music, keeping in touch with their friends and in co-listening. Participants reported that they are generally comfortable with co-listening to friends or being listened to and would use Colisten to see what their friends were doing or what mood they were in.

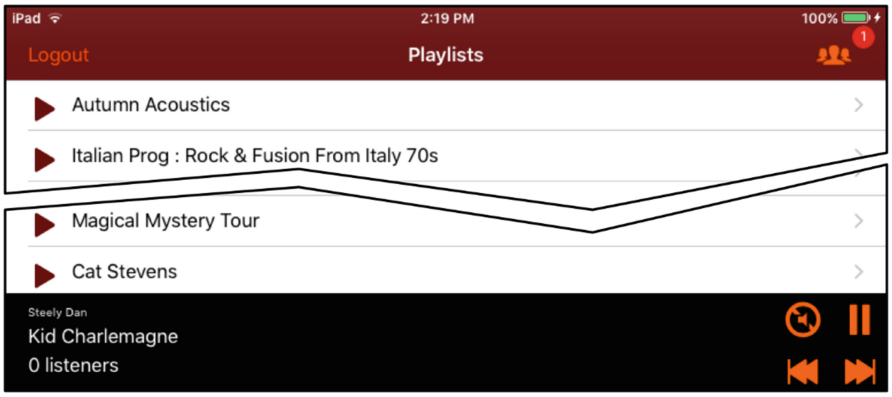


Figure 2.9 Colisten Interface (Stewart et al., 2018)

The presented studies all show the high potential of location-based music sharing applications. The tested applications were found to be entertaining and to stimulate curiosity. Additionally, they have been proven to foster connectedness and commonality and to promote social interactions. "The sharing and discovering of music [...] is perceived as an engaging activity resulting in piquing users curiosity about people in vicinity" (Seeburger et al., 2012). Based on the findings and design suggestions of these studies, we propose our own location-based music sharing application, SOOL. None of the examined studies uses our proposed combination of sharing with collocated strangers currently present in proximity, allowing for synchronous playback and using streaming services to provide full bilateral access to songs.

Chapter 3

Concept

In this chapter, we will present the findings from our literature review. The main goal of this thesis is to devise universal best practices and design paradigms for creating applications focused on improving mobile listening experiences, engaging the isolating effects of mobile listening practices and encouraging social interactions in public spaces. We analyzed studies on mobile music discovery, contextual music recommendations and technology-mediated social interaction to summarize their design choices and findings into reoccurring concepts.

3.1 Music as Connecting Element

The ability of music to connect people, encourage social interaction and bond over similar tastes makes it the perfect medium for applications aiming to create a sense of togetherness between people. As mentioned in section 2.1, music listeners self-identify through their tastes and are eager to self-express themselves by displaying them to other people. They therefore constitute an attractive user group for user interaction studies (Seeburger et al., 2012). When sharing music with other people, users will seek to form a connection to their audience by adapting their listening habits and choosing songs they feel represent themselves or they think other users will like (Voida et al., 2006; Seeburger et al., 2012; Kirk et al., 2016). Kirk et al. (2016) argue that "this accentuates the social nature of music listening, arguably reversing the criticism of private headphone 'cocooning'". Portable music can be used to connect - or simply associate - with other people as a common ground for interaction (Seeburger et al., 2012).

3.2 Sharing With Strangers

While existing location-based social networking services mainly focus on connecting friends, friends of friends or use profiling and matchmaking algorithms to connect users with similar

tastes, multiple studies have researched the aspect of anonymously sharing information with strangers in public spaces (Jacobsson et al., 2005; Bassoli et al., 2006; Seeburger et al., 2012; Kirk et al., 2016; Paasovaara et al., 2018). Participants were generally comfortable with sharing non-privacy data with strangers as long as anonymity was ensured. They also felt a lot of curiosity about the originators of some information and made guesses about the preferences of collocated people. The new potential for voyeurism engages people with their surroundings, raises interest in collocated people and creates a sense of cohesion (Seeburger et al., 2012; Paasovaara et al., 2018). "Even passive contact with strangers is perceived as rewarding and attractive when spending time in public urban places" (Seeburger et al., 2012). Participants generally have reacted very positively to this addition to their already existing habits: "I like it as it just adds this extra level of community and engagement to what you listening to which is nice. And it's fun because you are interacting with people around you" (Seeburger et al., 2012).

Using strangers as a source for music recommendation also has proven positive effects on serendipity and musical diversity (Bassoli et al., 2006; Seeburger, 2012; Åman et al., 2015; Kirk et al., 2016). "The thing I liked about it was that there was a lot of music I hadn't heard before" (Seeburger et al., 2012). Participants also speculated about the potential of public music sharing services to spawn localized music listening communities sharing interest for the same sub-genres.

3.3 Temporal Aspects

We could identify two different approaches concerning the temporal aspects of user interaction: Asynchronous and synchronous user interaction.

3.3.1 Asynchronous Interaction

Asynchronous interaction means users don't interact with each other in real-time. This is the case in systems, where users are free to leave digital artifacts that can be interacted with at any time that is not necessarily the present time. An example are forum posts that can be accessed, read or replied to at anytime a user visits the forum. Interaction may also be restricted to specific context (e.g. locations, time, weather), as further described in section 3.5.

Another example of asynchronous interaction is the creation and placing of digital artifacts as described in section 3.6.1.

Location-based studies showed that asynchronous interaction can serve as a sort of timeline for a certain place or environment and can give public places an additional layer of meaning (Åman et al., 2013). "Experiencing a place through time enriched the user experience through the (Gestalt theory's) principle of continuity" (Åman et al., 2015).

3.3.2 Synchronous Interaction

In contrast to asynchronous interaction, synchronous interaction is restricted to the present time and context (e.g. both users are online at the same time, both users are in the same location at the same time).

Location-based synchronous interaction is also called collocation. This principle is further described in section 3.6.2. Collocation has been proven to incline users to pay more attention to their surroundings. Users reported feeling more connected with collocated people and places when they knew other users were present.

3.3.3 Synchronous Playback

Only a few studies have implemented synchronous music playback between users (Bassoli et al., 2006; Stewart et al., 2018).

Bassoli et al. (2006) believe synchronization to be an important aspect of music experience and further explained: "[...] synchronized Music consumption among people in physical proximity, as it happens in clubs or during parties, can create a strong emotional connection, more than asynchronous download of music over distance could provide [...]"

Stewart et al. (2018) argue that "the lack of co-listening is due to design gaps in sharing features rather than lack of interest".

3.4 Communication

The examined studies showed different communication concepts. Some studies completely restricted communication to exchanging music. This aims to keep interaction anonymous and superficial to emphasize the social agency of music. Other studies implemented various communication mechanics that can be grouped into indirect and direct communication.

3.4.1 Indirect

We define indirect communication as mechanics that allow users to react to each other through predefined restricted methods that allow for no personal amendments. Example mechanics are the giving of 'Likes' to users or songs or automated notifications signaling nearby users or that a user has just viewed another user's profile or song. Leaving comments to encountered musical choices allow users to sympathize with other users and exchange some personal information like lifestyle, humor or creative associations (Åman et al., 2015). Studies that implemented such

communication report that even such limited communication can have great impact on perceived closeness or connectedness (Seeburger et al., 2012; Åman et al., 2015).

3.4.2 Direct

We examined multiple studies using direct communication in the form of chats (Bassoli et al., 2006; Seeburger et al., 2012) or annotations (Kirk et al., 2016; Paasovaara et al., 2018). Paasovaara et al. (2018) studied the aptitude of PNAs to create social interaction in detail, aiming to connect collocated strangers through messages to the point of face-to-face meetings. They reported that direct technology-mediated interaction has the aptitude for fostering real-life social interaction. As mentioned in section 2.1, music has an inherent ability to connect people. Seeburger et al. (2012) reported that participants were generally comfortable with establishing direct contact with strangers sharing the same taste in music. They argue that this can lead to meaningful technology-mediated interactions.

3.5 Positive Restrictions and Context Sensible Services

The modern ubiquity of mobile internet provides owners of mobile devices with continuous access to a multitude of applications, functions, internet resources and other services. This enables users to use those services in a great variety of different contexts (Alarcón et al., 2006). Context sensible services are services that are only available if certain conditions are met. Often employed contexts include certain time windows (e.g. limited events, time of day), locations (e.g. when the user is in the proximity of a certain place or other users), network access (e.g. iTunes, where file sharing is limited to users in the same subnet), activities (e.g. music recommendation based on activity) and weather (e.g. in Pokémon Go, encounter chance and strength of certain Pokémon types are dependent on the weather conditions) (Wang et al., 2012).

Studies showed that the quality of serendipitous encounters was higher with content that could not be ubiquitously accessed, making them more interesting and exciting (Voida et al., 2006; Seeburger et al., 2012; Åman et al., 2015; Paasovaara et al., 2018). Limiting contextual factors that foster appealing novel interaction experiences are called 'positive restrictions' (Åman et al., 2015).

3.6 Location Context

Location-based approaches show especially promising results in connecting users of mobile devices to their surroundings (Bassoli et al., 2006; Baumann et al., 2007; Tanaka et al., 2007; Seeburger

et al., 2012; Åman et al., 2015; Kirk et al., 2016). In location-based applications, the concepts of synchronous and asynchronous interaction (see section 3.3) take the form of leaving artifacts at specific locations and interacting with collocated users in real-time.

3.6.1 Leaving artifacts/information

In the field of mobile music listening, this concept has often been employed using location as a context. Multiple systems allow for users to drop musical recommendations as digital artifacts at specific locations (Åman et al., 2015; Kirk et al., 2016). These artifacts can be interacted with by other users when they are within a predefined reach of this location. Some systems allow for artifacts to be placed freely in the world, while others restrict the placement of artifacts to locations like landmarks or predefined locations.

Multiple studies allow users to asynchronously share music recommendations, song files or other virtual objects by placing them at specific landmarks, predefined places or at self-chosen locations (Åman et al., 2015; Kirk et al., 2016). Those virtual artifacts can then be seen, listened to, collected or commented on by other users.

Being able to encounter artifacts stimulates the users' curiosity and ambition to encounter even more artifacts. Searching for and collecting artifacts is often perceived as amusing and rewarding: "Participants enjoyed the ability to spontaneously scan their environments to find entertainment for free time" (Åman et al., 2015).

3.6.2 Collocation

Users are defined as collocated when they are in physical proximity to either another or the same place at the same time. In some studies, predefined locations serve as a kind of 'virtual chatroom' anchored in the real world, where users can interact in real-time. We examined multiple studies researching the effects of real-time interaction with collocated users (Jacobsson et al., 2005; Bassoli et al., 2006; Baumann et al., 2007; Seeburger et al., 2012; Paasovaara et al., 2018). Seeburger et al. (2012) found that collocation "stimulates curiosity and voyeurism". They reported that participants were periodically scanning their surroundings, searching for other users. Paasovaara et al. (2018) reported that their participants showed a heightened awareness of collocated strangers and increased curiosity towards their surroundings.

3.7 Technology

In this section, we discuss examined design principles concerning technical aspects.

3.7.1 State-of-the-Art

As described in section 2.1.1, electronic end-user products are fast changing in functions and capabilities. Old technologies are often replaced by improved successors, making them less attractive, if not obsolete. To guarantee a system's long-term relevance, researchers should always aim to adopt new technologies and build upon its features (Tanaka et al., 2007; Kirk et al., 2016). Modern smartphones are constantly improving in terms of performance, storage, connection speeds and additional sensors allowing for the development of novel, innovative and more sophisticated applications.

3.7.2 Popular Technologies

When designing for widespread popularity, researchers should analyze the technologies currently being used by their target user group and their respective pervasiveness. Systems should then be developed for the most popular and widely used technologies to ensure widespread adaption (Watson et al., 2016).

A good negative example for this is the study by Bassoli et al. (2006). They developed their system 'tunA' for Personal Digital Assistant (PDA) devices. None of their participants had ever used a PDA before, and their system gained some criticism for their choice of platform. One participant remarked; "I really liked the interface with the touch screen, I think its just a fact that if I had a PDA I'd feel that I had to use it for more than just the tunA thing, so I would much prefer to pay half the price and get a normal portable player" (Bassoli et al., 2006).

3.7.3 Streaming Music

We encountered many studies that did not employ actual music sharing but rather sharing of song choice metadata (Seeburger et al., 2012; Baumann et al., 2007) or short song samples (Kirk et al., 2016).

Some examined systems allowed for the sharing of MP3 files with collocated users. Bassoli et al. (2006) and Jacobsson et al. (2005) enabled sharing with collocated strangers, Tanaka et al. (2007) only with collocated friends.

We argue that nowadays the sharing of MP3 files often constitutes a copyright violation. It further limits the possible song choices to music the users actually possess and have saved on their device.

The proliferation of modern streaming services solves these problems. Spotify, being to most popular streaming platform (see section 2.1.3), enables free access to their vast collection of music to any user with an account and internet access. Given the ubiquitous access to vast musical libraries and their popularity and pervasiveness, we advise using streaming services as a source for mobile music sharing applications.

3.7.4 Location Technologies

There is some controversy about the usage of either Bluetooth or Global Positioning System (GPS) when designing PNAs.

Using Bluetooth as indication for physical proximity is a straightforward approach - devices are either in range or are not. Most modern mobile devices use Bluetooth transmitters that have an approximate range of 10 meters. When a connection between both devices can be made, collocation is assured. Still, even in modern times, Bluetooth tends to be unreliable. It has fluctuating ranges for it is easily blocked by physical barriers like walls, doors, metal or bodies and is also vulnerable to electromagnetic disturbances.

Paasovaara et al. (2018) discourage the use of Bluetooth for early-stage PNAs when a sufficient user density cannot be guaranteed. In their field trial, this led to a discouraging scarcity of encounters and, therefore, disappointing results. Instead, they rather advertised the use of GPS and adjustable ranges to accommodate for lack of user density as needed. Also, Bluetooth only enables location-awareness to other Bluetooth devices and not to specific locations like addresses, landmarks or self- and predefined locations.

In PNAs, Bluetooth simultaneously handles location and data transmission. When using GPS, data transmission has to be realized through other means (e.g. over mobile internet using a server). With today's ubiquitous access to fast mobile internet and the pervasiveness of mobile devices with GPS functionality, we still judge the employment of GPS location to be the better option.

Collocation can also be determined by network connection (e.g. iTunes' file sharing option only being available to users being connected to the same subnet). This option may also be feasible for select use cases.

Chapter 4

Design and Implementation

In this chapter, we will summarize the implemented concepts, design choices, architecture and technologies used to create SOOL.

4.1 Concepts

SOOL is a real-time location-based music sharing application using state-of-the-art technology. It implements the following concepts:

- music sharing
- sharing with strangers
- location-based sharing
- asynchronous interaction
- collocation
- using popular state-of-the-art technology (smartphones, mobile data, Spotify streaming service)
- synchronous listening

Similar to Seeburger et al., 2012; Kirk et al., 2016 and Åman et al., 2015, our first prototype did not implement all features the final system is desired to incorporate. The following features are planned to be included in future iterations:

- direct and indirect communication (chat, likes and notifications)
- asynchronous sharing (free placement of song artifacts)
- user-profiles and in-app song collections

4.2 Installation

Installation of the app had to be realized differently for Android and iOS. Android allows for the installation of APK files which we generated using the development framework Expo. iOS does not allow for custom apps to be installed on devices.

4.2.1 Android

On Android, the app can be installed using the provided APK files. 'sool.apk' is the full version of the current iteration, while 'sool_test.apk' was slightly modified for the user study. Differences in the study version include the removal of yet still non-functional User Interface (UI) elements and adaptation for iOS-support.

4.2.2 iOS

Apple does not allow for free app development. For the user study, we used the development and distribution framework Expo running on a PC to distribute a development version. To run the app on iOS devices, the PC has to have Node.js installed. The project is located in the folder 'Expo Project'. Running 'npm install' in the command-line interface (CLI) will install all dependencies as well as the Expo framework on the machine. 'npm run start' will build the app and create a QR code that can be scanned on iOS devices using the 'Expo Go' application that is available on the App Store.

4.3 UI Design

The visual elements and interaction patterns of SOOL were designed to closely resemble those of the mobile or desktop application of Spotify (see figure 4.4).

4.3.1 Logo

Figure 4.1 shows the provisional logo design for SOOL. It uses the cell tower metaphor with Spotify logos as radio waves to signify the broadcasting of music.

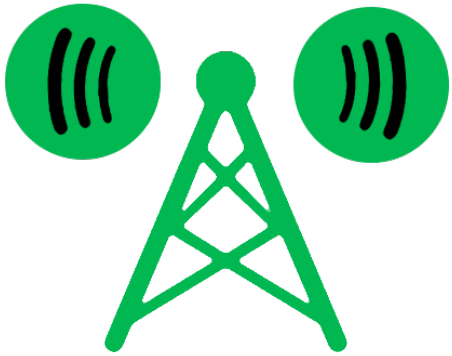


Figure 4.1 SOOL Logo

4.3.2 Colors

SOOL uses the same color palette present in the Spotify application. The green, white and black hex values are defined by Spotify for Developers (*Spotify — Company Info* 2021). We extracted the hex values for anthracite, dark grey and light grey from the Spotify desktop application as well as adapted the official black slightly to our design (see figure 4.2).

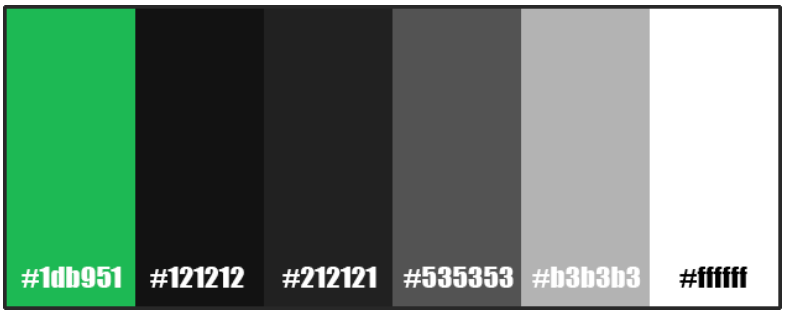


Figure 4.2 Spotify color palette

4.3.3 Font

Spotify developed its own licensed font called Spotify Circular. We chose a freely available font closely resembling that used in Spotify's design called Proxima Nova Bold (*Proxima Nova | Adobe Fonts* 2021) (see figure 4.3).



Figure 4.3 Proxima Nova Bold Font (*Proxima Nova | Adobe Fonts* 2021)

4.3.4 UI Elements

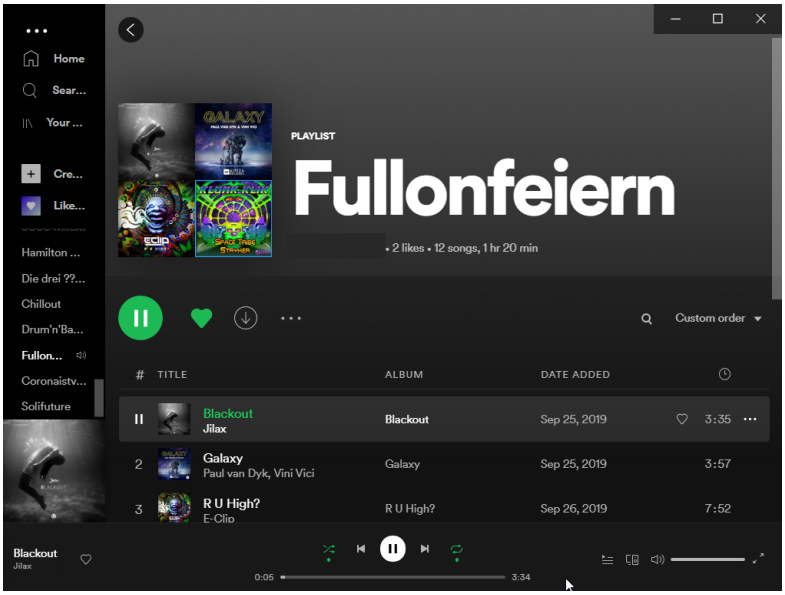


Figure 4.4 Spotify desktop application UI

Bottom Bar

Similar to the Spotify desktop client, SOOL has a bottom bar providing access to key features of the application (see figures 4.6 to 4.11). In Spotify, the bottom bar consists of a display for the currently playing song on the left side, controls for the playback in the centre and access to the current queue and volume controls on the right side. To create interaction analogies, we arranged the bottom bar in a similar fashion, displaying the current song on the left side, a button to toggle sharing in the middle and a button leading to a list of currently collocated users on the right side. Another button on the right side leads to a list of users currently listening to one's own song choices. As this feature is not yet implemented, the button has been removed for the user study.

Top Bar

The top bar displays the current navigation route's name, a home button (when not currently on the 'Home' screen) and a button showing unread notifications that navigates to the notification screen when clicked and a button leading to the app's settings.

Buttons

The designs of the buttons are distinct representations of their respective functions to create a UI that is intuitively understandable. We chose visual metaphors that are either already well established in the field of UI design or that are easy to interpret (see figure 4.5). Like in the Spotify application, active buttons are highlighted in green, inactive buttons in white.

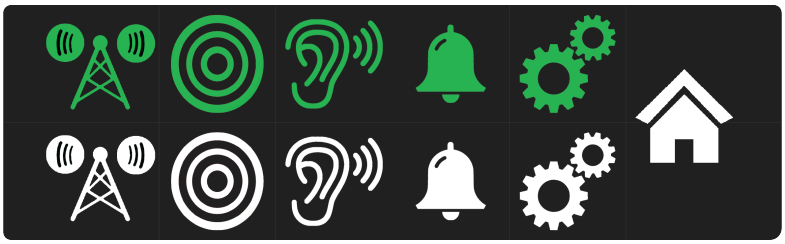


Figure 4.5 SOOL Buttons

- Sharing playback is represented by a broadcasting cell tower symbol employing two Spotify logos as radio waves.
- Nearby users are represented by multiple rings inside another meant to resemble a radar screen.
- Listeners are represented by an ear with sound waves.
- Notifications are represented by a bell.
- Settings are represented by gears.
- The 'Home' button is represented by a house.

4.3.5 Screens

SOOL is divided into different screens for its different features. Users can navigate between the screens using UI buttons. In the following, we will list all screens and their content.

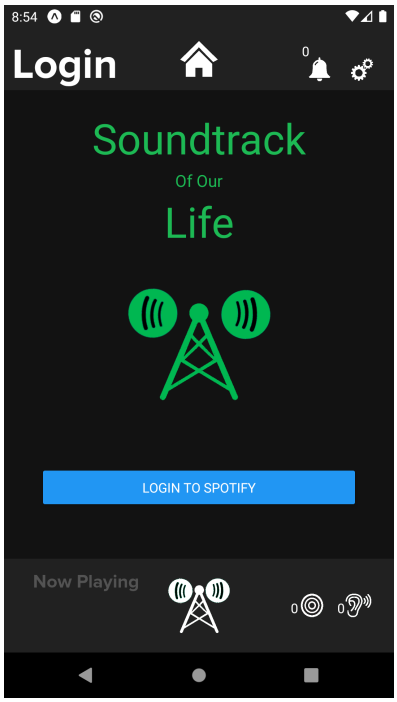


Figure 4.6 Login Screen

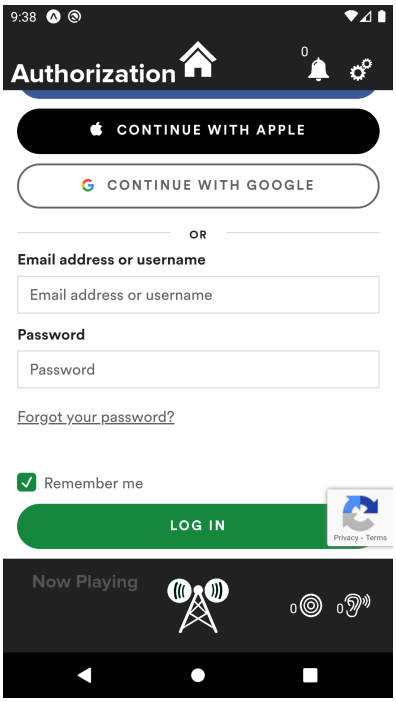


Figure 4.7 Authorization Screen

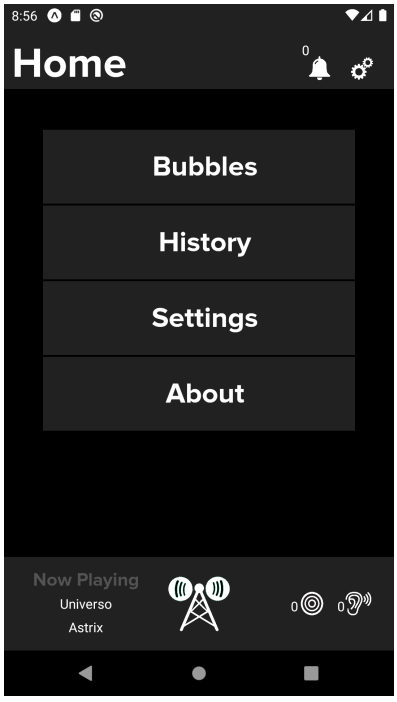


Figure 4.8 Home Screen



Figure 4.9 Listen Screen

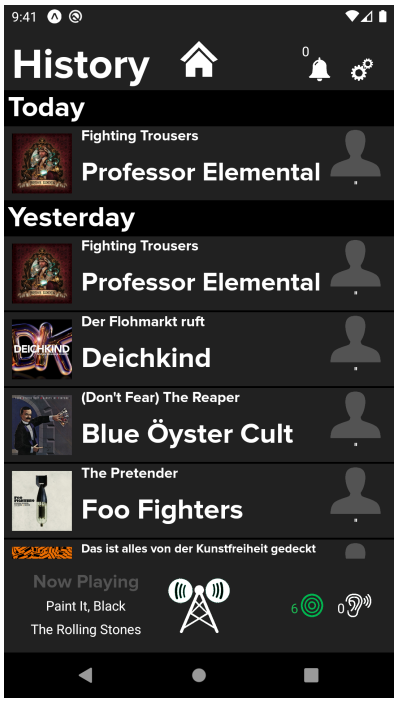


Figure 4.10 History Screen

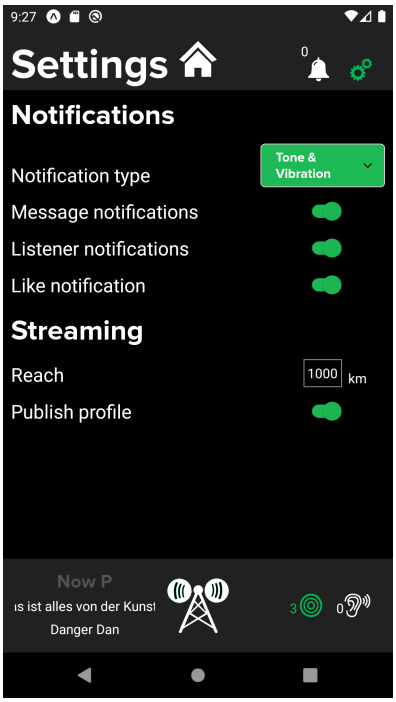


Figure 4.11 Settings Screen

Login Screen (see figure 4.6)

The 'Login' screen is SOOL's first in-app screen. The screen displays SOOL's name and Logo as well as a 'Login to Spotify' button, which navigates to the 'Authorization' screen. The 'Login' screen will check if a user is already authorized, in which case it will automatically redirect to the 'Home' screen.

Authorization Screen (see figure 4.7)

The 'Authorization' screen provides a website served by Spotify's Web API, allowing the user to login into their Spotify account. Once that is done, the user can leave the screen via the systems back button or the 'Home' button in the top bar.

Home Screen (see figure 4.8)

The 'Home' screen offers a menu with buttons navigating to 4 other screens: 'Listen' screen, 'History' screen, 'Settings' screen and 'About' screen. The 'Listen' screen is equally reachable through the buttons of the bottom bar as is the 'Settings' screen through the top bar.

Listen Screen (see figure 4.9)

The 'Listen' screen displays a scrollable list of all other users that are in reach and are sharing songs. The list entries can be clicked and will then open the song in the user's active Spotify application. If the listed users' profile pictures are clicked, that users' Spotify profile page will be opened in the Spotify application instead.

History Screen (see figure 4.10)

The 'History' screen displays a scrollable list of all songs a user has shared through SOOL. Entries are sorted by time and categorized into 'Today', 'Yesterday', 'This week', 'This month', 'This year' and 'Older'.

Settings Screen (see figure 4.11)

The 'Settings' screen offers the possibility to choose notification types, turn off notifications, define the reach of SOOL and whether or not the user chooses to publish his or her profile when sharing. In the current version, only the adjustment of reach has been implemented.

4.4 Example of Use

This section will demonstrate a typical usage flow for using SOOL with the currently implemented features. Our example user is named Alex and has freshly installed SOOL.

Alex is currently riding the tram. Alex starts the app. On the 'Login' screen Alex presses the button 'Login to Spotify'. SOOL navigates to the 'Authorization' screen. Alex enters the Spotify credentials and clicks on 'Login'. SOOL signals a successful login and prompts Alex to go back. Alex navigates back to the 'Login' screen, which in turn redirects to the 'Home' screen. The bottom bar's 'Listen' icon will display whether and how many other users in reach are currently sharing songs. Alex navigates to the 'Listen' screen by clicking either the 'Listen' menu item or the 'Listen' icon in the bottom bar. Alex is presented with a list of songs shared by other users. Alex clicks a song from the list. If Alex' device has the Spotify app running, that song will be started in the background at the timestamp of the sharing user's playback. If Spotify was not running before, it would now be opened. Alex listens to the song for some time, then clicks on another song. Alex looks around and makes assumptions about who might be sharing what. Alex repeats this a couple of times with different songs. Alex then decides to share music. Alex switches to the Spotify app and picks a song that Alex likes a lot. Alex switches back to SOOL. Alex' current playback is displayed in the bottom left corner, so Alex knows that SOOL

has correctly recognized the song. Alex then presses the sharing icon in the middle of the bottom bar, which in turn lights up green to signal that sharing is active. Alex continues to share for the duration of the tram ride, then, when arriving at the final stop, gets off the tram and stops sharing.

4.5 Architecture and Technology

This section will discuss SOOL's architecture and the technologies used during development.

We developed SOOL for Android and Apple iOS Devices using the React Native framework in combination with the Expo development framework. The SOOL client app requests data from Spotify's API about the current playback of a user using a Node.js backend server. The SOOL client app connects to the database server to store and connect collocated users and to save the users' sharing history and log data. Communication with the database as well as filtering of nearby users is realized through another Node.js backend server. The servers are hosted using Heroku, a cloud platform as a service that handles the deployment of Node.js web servers via the Git version control system.

SOOL's architecture is visualized in figure 4.12.

4.5.1 React Native

React Native works similar to the React framework, using a structure resembling HTML/CSS with JavaScript to build user interfaces. React Native additionally allows apps to directly communicate with the native platforms, enabling seamless cross-platform app development.

4.5.2 Database

For our prototype, we chose MongoDB, a document database, for it's ease of use and because it is read-optimized for user entries and histories which is the primary query. The MongoDB database is hosted by MongoDB Atlas, a cloud database service by the MongoDB developers that takes charge of the deployment and management of the database. The SOOL client accesses the database using another Node.js backend server with the Express.js framework that connects to the database using the 'Mongoose' JavaScript library.

The database has 3 collections. 'users', 'histories' and 'logs'.

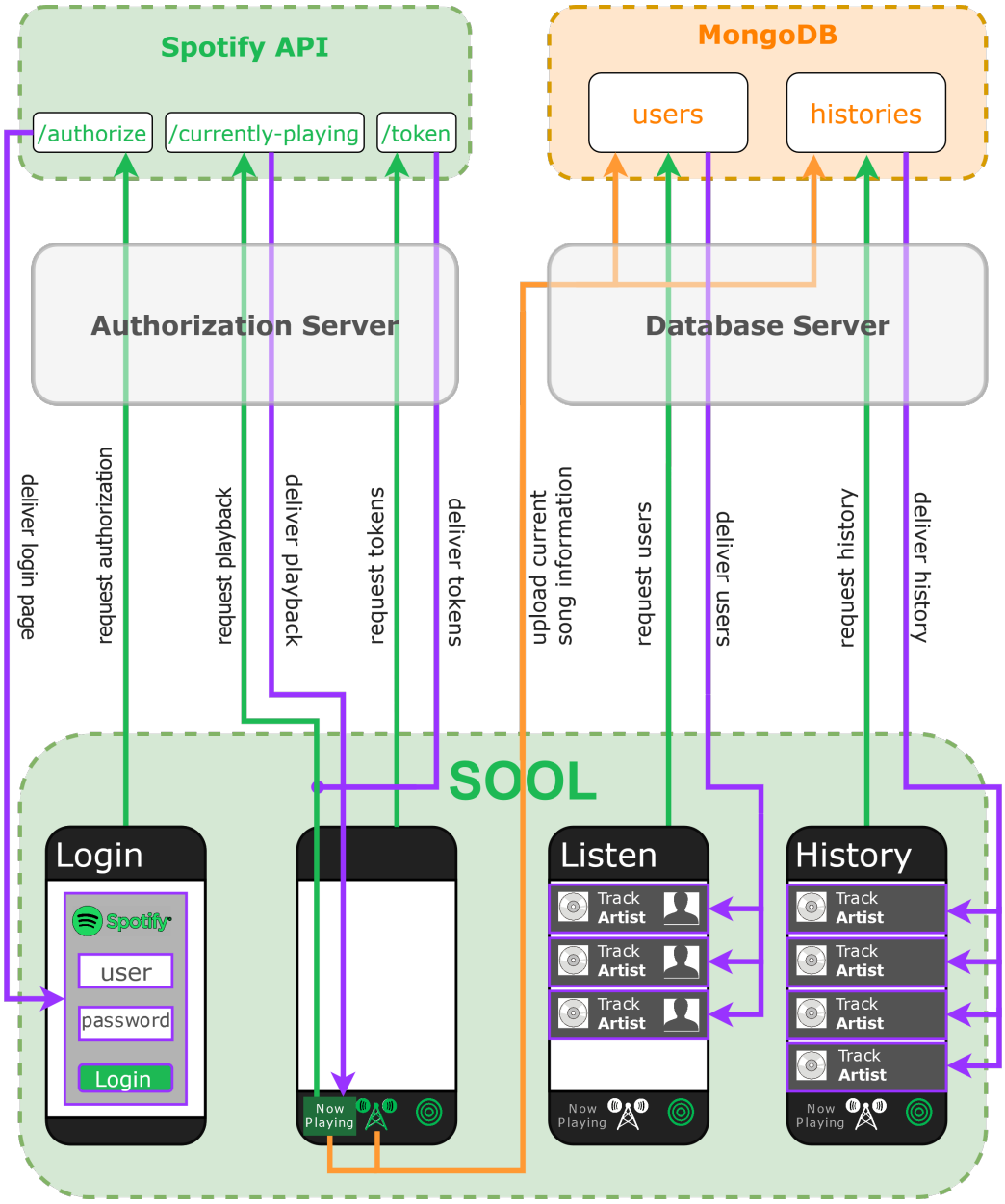


Figure 4.12 SOOL Architecture

Users

The 'users' collection stores user-song combinations of users currently sharing their playback. A 'User' entry has the following fields:

- **location** (Object) - a GeoJSON object holding the user's coordinates
- **userId** (String) - the user's Spotify id
- **userPictureUrl** (String) - a String holding the Uniform Resource Locator (URL) to the user's Spotify profile picture
- **song** (Object) - an Object holding various information about the user's song choice and playback
  - **title** (String) - song title
  - **artist** (String) - song artist
  - **progress** (Number) - current playback timestamp in milliseconds
  - **duration** (Number) - total song duration in milliseconds
  - **spotifyUrl** (String) - link that opens the song in Spotify web page
  - **spotifyUri** (String) - link that opens song directly in Spotify app
  - **imageUrl** (String) - link to song or album cover image
- **lastUpdated** (Date) - time of last update to database used to deal with connection losses
- **createdAt** (Date) - time of entry creation

Histories

The 'histories' collection stores user-song combinations of songs users have shared. A 'History' entry is derived from a 'User' entry but has fewer fields. It has the following fields:

- **location** (Object) - a GeoJSON object holding the user's coordinates
- **userId** (String) - the user's Spotify id
- **song** (Object) - an Object holding various information about the user's song choice and playback
  - **title** (String) - song title
  - **artist** (String) - song artist
  - **progress** (Number) - current playback timestamp in milliseconds
  - **spotifyUri** (String) - link that opens song directly in Spotify app
  - **imageUrl** (String) - link to song or album cover image
- **createdAt** (Date) - time of entry creation

Logs

The 'logs' collection stores various usage statistics monitored by the app. A 'Log' entry has the following fields:

- **userId** (String) - the user's Spotify id
- **appOpenedCount** (Number) - How often the app was started?
- **appTimes** (Array) - app usage durations in milliseconds
- **sharingTimes** (Array) - sharing durations in milliseconds
- **listeningCount** (Number) - How often did the user open another user's song choice?
- **listeningTimes** (Array) - How long did the user listen to other user's songs?
- **createdAt** (Date) - time of entry creation

'appTimes' and 'listeningTimes' have not yet been implemented and are planned to be included in future iterations of SOOL.

When a user starts sharing, the SOOL client sends information about the user's Spotify account, the user's location and the user's song choice to the database server. The database server then creates entries in the 'users' collection and in the 'histories' collection. While sharing, the respective entries will be compared to the current playback every 5 seconds. If the song has changed, the 'User' entry will be updated and a new 'History' entry will be created. When the user stops sharing, the user's entry will be removed from the 'users' collection.

To find nearby users, the SOOL client sends a request to the database server containing the user's current location and the current reach set in the app. The server then retrieves all currently sharing users from the 'users' collection and calculates their distance to the user using the haversine formula. The results are then filtered by the user's reach setting and sent to SOOL in JavaScript Object Notation (JSON) format.

This approach does not scale since in the current implementation it would follow O(n). This is feasible only for small user bases as present in our prototype user study. For future implementations we plan to use a database plugin or a different database for native geospatial search.

When the SOOL client requests a user's history, the database backend server retrieves all history entries from the 'histories' collection selected by the user's id and sends back the entries in JSON format.

The SOOL client logs different usage events or durations throughout the app. These events and durations are sent to the database server, which in turn increments the counts of events or adds durations to the respective arrays. The duration arrays' sizes also serve to determine the respective counts.

Information is updated using the short polling principle, which sends repeated requests in short intervals to the database, updating entries or retrieving up-to-date collections every 5 seconds.

4.5.3 Spotify API

Spotify's API is based on REST principles and can be accessed via HTTPS requests. Responses usually deliver JSON data containing user data. In order to access Spotify's API, permissions have to be granted first through the 'Spotify Authorization Flow', which implements the OAuth 2.0 authorization framework (see figure 4.13). SOOL sends requests to the Spotify API through a Node.js backend server using the Express.js framework that handles the requests to the API's endpoints.

First, the app has to be registered at Spotify for Developers to get credentials for the app and to define a static redirect URL for user authorization. The credentials can be used to access Spotify API's authorization endpoint, which in turn displays the scope of the app's requested excess to the user's Spotify profile and prompts the user to login and authorize access. Upon completion, the Spotify API returns a code generated from the apps client id and the user's Spotify id. This code combined with the app's credentials can then be used to request user-specific access and refresh tokens which will be sent to the previously defined redirect URL. Access tokens can, in turn, be used to access the Spotify API's endpoints that are within the previously defined scope. SOOL requests access to the scopes *user-read-private* and *user-read-email* to access the users' profile information and to the *user-read-playback-state* scope to access the users' currently playing song. Access tokens are usually valid for one hour, after which a new access token can be requested using the refresh token.

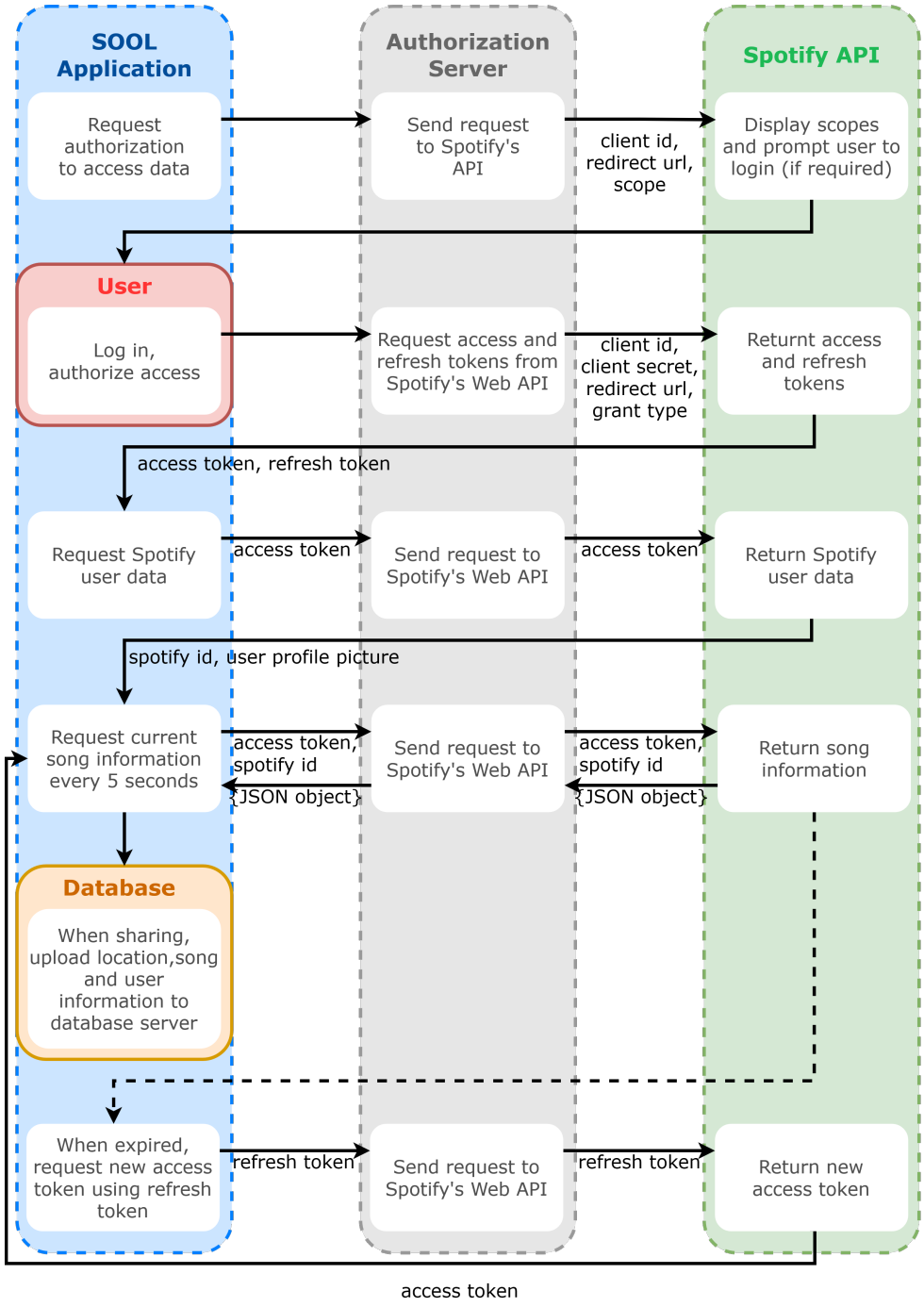


Figure 4.13 SOOL's Spotify Authorization Flow

Chapter 5

User Study

We organized a user study to confirm our overall design and to gain feedback about the aptitude of the implemented concept for improving listening experiences and fostering social interaction. Additionally, we aimed to encourage reflection about social aspects of mobile music consumption among our study participants. Our research questions were: *Is the app engaging and fun to use? Does the proposed concept provide novel forms of interaction and foster more connectedness among users? Does the concept provide unique means to escape the musical filter bubble and discover new music?*

To answer these questions, we conducted a user study with 7 participants over one hour. Our aim was to assess the overall usability of our prototype as well as to gain qualitative feedback about the employed concepts and design choices. For that we gathered log data during the experiment and conducted interviews and used structured questionnaires. We also conducted a survey on the participants' music listening habits to later contrast them with the results of our study.

5.1 Participants

We recruited 7 participants (3 male, 3 female, 1 preferred not to say) between the ages of 20 and 28 years (M= 24 years) from a shared apartment with a total of 15 inhabitants. The participants knew each other at varying degrees of intimacy, some having lived years together, some only several weeks. 6 participants were students, and 1 participant was employed full-time. Participants were recruited through a group chat and invited to the introductory group session in the dining hall. As per the requirement for using the Spotify API, only Spotify Premium users could take part in the study. One participant reported that he usually doesn't use Spotify and was provided with a Spotify premium account for the duration of the study. Participants were reimbursed with a tasty self-cooked meal.

5.2 Procedure

Preparation and Testing

Participants were provided with a download link for the application file and received instructions and help for installing it on their devices. For Apple devices, the application was installed using the 'Expo Go' app, which is used to build React Native apps that are still in development. The participants were then explained how SOOL works and how to use its main functions. These included sharing a song with users in your vicinity, searching for collocated users and tuning into their playback, viewing a user's Spotify account and accessing your playback history. The participants were then asked to try out sharing a song and to tune in to the playback of another participant. After that, participants were free to use the app to their will.

Free Usage Phase

In the following free usage phase, we collected in-app usage data, which included:

- times app was opened
- times turned on sharing
- sharing durations
- times turning into some else's playback

This data was later used to provide a statistical usage overview and evaluate user behavior.

Debriefing

At the end of the study, the participants were individually debriefed. Each participant had to fill out a number of questionnaires as well as partake in a semi-structured interview.

5.3 Questionnaires

Inclusion of Others in the Self Scale

Before and after the experiment, the participants evaluated the felt closeness to the other participants using the Inclusion of Others in Self Scale (IOS Scale) by Aron et al. (1992). The IOS Scale is a single item questionnaire where subjects have to choose the pair of circles best describing their relationship to another person (see figure 5.1). It is deemed a "psychologically meaningful and highly reliable measure of the subjective closeness of relationships" (Gächter et al., 2015). For this, participants were given an online form where they could pick an alias to avoid biases.

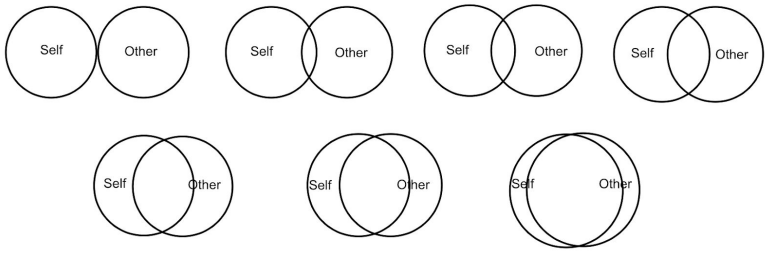


Figure 5.1 Inclusion of Others in Self Scale (IOS Scale) (Gächter et al., 2015)

Demographics and Music Listening Preferences

Each participant had to fill out a demographics questionnaire (Table: A.1.1) asking for age, gender and occupation. Afterwards, they were presented with a music listening preferences questionnaire to provide some background information about the participants' habits (Table: A.1.2). The questionnaire investigated their practices of music consumption with a focus on public environments (e.g. How often do you listen to music on Spotify? (Q1), How do you discover new music? (Q3), Do you ever wonder what other people are listening to in public environments? (Q7), Why do you listen to music in public environments? (Q8)).

AttrakDiff Questionnaire

To get quantifiable data about the quality of the app, we employed the 'AttrakDiff' questionnaire (Hassenzahl et al., 2003) as a quick and reliable way to analyze the user experience (UX). By confronting the user with 28 bipolar word pairs on a 7-point Likert scale (e.g. simple-complicated, predictable-unpredictable, innovative-conservative (see table A.1.3)), the questionnaire is able to reliably assess the pragmatic and hedonic qualities of a product as well as its attractiveness (Hassenzahl et al., 2003).

5.4 Interviews

The following semi-structured interview consisted of a set of five questions aimed to instigate qualitative comments concerning the concept of sharing music with collocated people using SOOL and Spotify. Interviews were recorded for later transcription and analysis.

Interview question catalogue:

- How did you like the app?

- What were your experiences when sharing/listening?
- What song choices did you choose to share? Why?
- Did you feel more connected to your flatmates using the app?
- Did you start talking or chatting about the song choices displayed in the app?
- Do you think SOOL can help you discover new music?
- Would you use SOOL? Why?
- Do you have improvements or suggestions?

Chapter 6

Results

This chapter presents and analyzes the results gathered through our user study as described in the previous chapter.

Participants' names and aliases have been replaced by numbers.

6.1 Music Listening Questionnaire

The music listening questionnaire shed some light on the participant's practices of music consumption.

The participants reported that on average they use Spotify between 5 to 6 days a week (see figure 6.1) (M=5.571, SD=1.618), spending about 2 hours per day listening to music on Spotify (see figure 6.2) (M=2.071, SD=0.976) (see table 6.1).

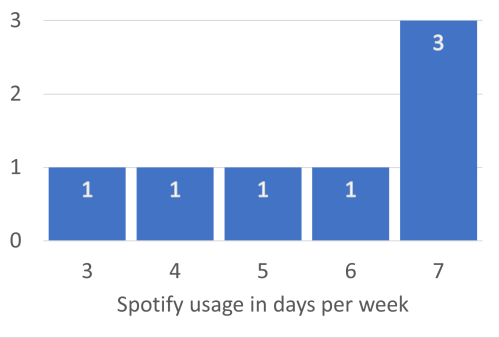


Figure 6.1 Q1: How many days a week do you listen to Spotify?

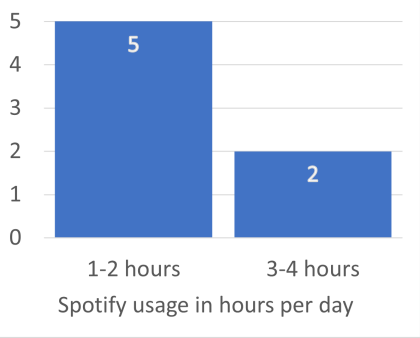


Figure 6.2 Q2: How long do you listen to Spotify every day?

When asked about their typical way of listening to music, all but one participant reported that

they usually streamed from their own playlist. 5 participants reported that they also streamed either playlists made by others or complete albums. 4 reported that additionally, they stream singular songs. 3 participants reported that they often visit concerts, 2 also listen to the radio. 1 participant reported listening to purchased digital music, and one participant reported listening to automatically created playlists (see figure 6.3).

To the question of how they mainly discovered new music 4 participants answered that friend's recommendations were their biggest source of new input, while 3 answered that playlists and mixes provided by streaming services were most influential to them (see figure 6.4).

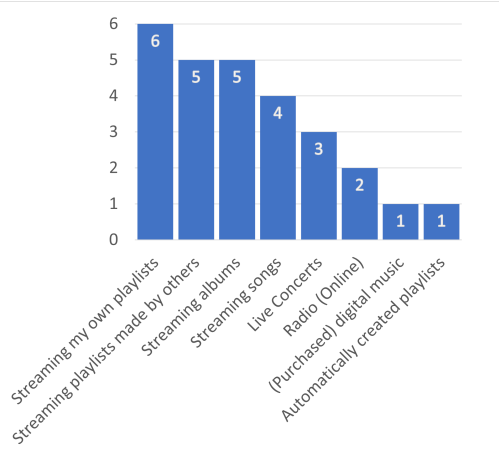


Figure 6.3 Q1: How many days a week do you listen to Spotify?



Figure 6.4 Q4: How do you mainly discover new music?

The participants reported that they are positively inclined to listen to music when alone in public places scoring an average of 4 out of 7 points on a Likert scale (see figure 6.5) (M=4.000, SD=1.528).

Participants were also positively interested in what other people were listening to in public spaces, scoring an average of 4.4 out of 7 points (see figure 6.6) (M=4.429, SD=0.976).

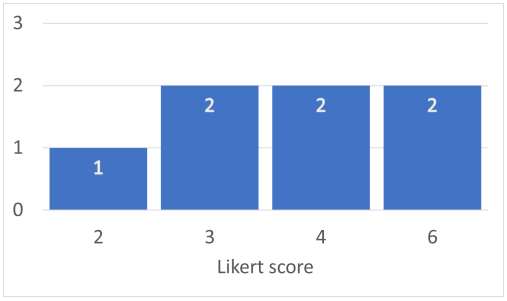


Figure 6.5 Q5: How likely are you to listen to music when alone in public spaces?

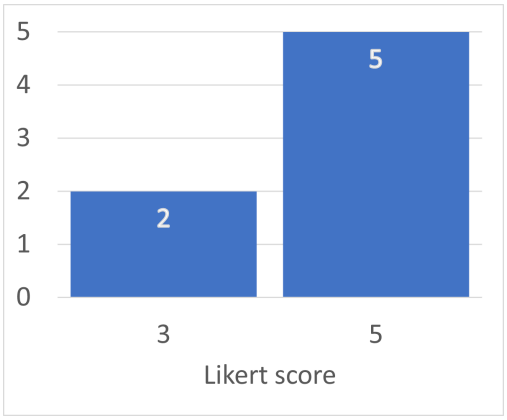


Figure 6.6 Q6: Do you ever wonder, what other people are listening to in public spaces?

As a final question, we asked the participants what their motivation is for listening to music in public spaces. 6 participants reported that relaxation and entertainment were the main reasons for listening to music in public. 2 participants also mentioned distraction as a factor, and one participant also named preparation (see figure: 6.7).

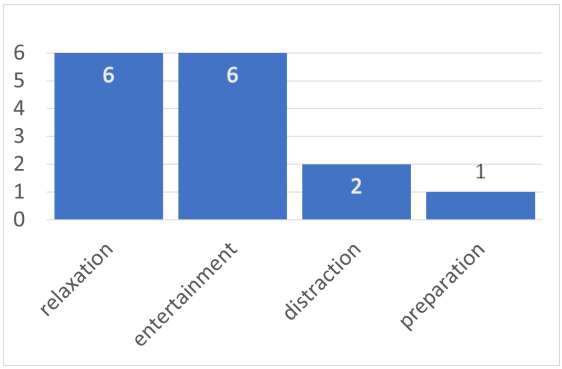


Figure 6.7 Q7: Why do you listen to music in public spaces?

Table 6.1 Music Listening Statistics

	Mean	Std. Deviation	Minimum	Maximum
Q1: How many days a week do you listen to Spotify?	5.571	1.618	3.000	7.000
Q2: How long do you listen to Spotify every day?	2.071	0.976	1.500	3.500
Q5: How likely are you to listen to music when alone in public spaces?	4.000	1.528	2.000	6.000
Q6: Do you ever wonder what other people are listening to in public spaces?	4.429	0.976	3.000	5.000

These results coincide with the findings of the preliminary literature study and confirm the participants as fitting subjects to our user study.

6.2 Usage Activity

Due to unforeseen errors during the study, the log and history data that we gathered was incomplete. The raw data misses some user's counts and times for sharing. This was very likely due to the study's unique and untested usage conditions. The number of users simultaneously accessing the app, as well as the integration of Apple users via the 'Expo Go' app, was only tested during the study. The resulting errors in logging the usage activity will be fixed in future iterations.

The following analysis was conducted for the sake of thoroughness. When viewed, it should be considered that actual values are likely to be significantly higher. The usage activity has therefore been excluded from our evaluation of SOOL.

The analysis of the logs fostered the following usage statistics: Over the duration of the study, participants on average opened the app 3.286 times (SD=2.812), tuned into others users' playback 5.429 times (SD=2.299) and turned on sharing about 2.286 songs (SD=2.870). On average, each sharing lasted nearly 3 minutes (M=2.887, SD=3.511). Over the course of the study, a total of 75 songs were shared. 23 of these songs have been re-shared when the user tuning in also had sharing enabled, leaving 52 genuine song choices or 7.4 songs per participant (SD=4.826).

6.3 AttrakDiff User Experience Assessment

Modern research in the field of HCI concluded that usability is not a sufficient measurement when it comes to analyzing the qualities and aptitude of a product (Hassenzahl et al., 2008). "Research has shown, that perceived enjoyment and perceived ease of use are stronger determinants towards technology adoption than perceived usefulness when evaluating hedonistic information systems" (Seeburger et al., 2012). In order to get an extensive overview of the UX, the AttrakDiff questionnaire analyzes the participant's subjective pragmatic and hedonic impressions (divided into stimulation and identity) as well as the felt attractiveness of the product. The score for pragmatic qualities shows the effectiveness and efficiency of the app. Hedonic stimulation rates the aptitude of the product to satisfy the user's longing for exploration and self-improvement. Hedonic identity rates the products aptitude for forming interpersonal relationships and displaying self-worth.

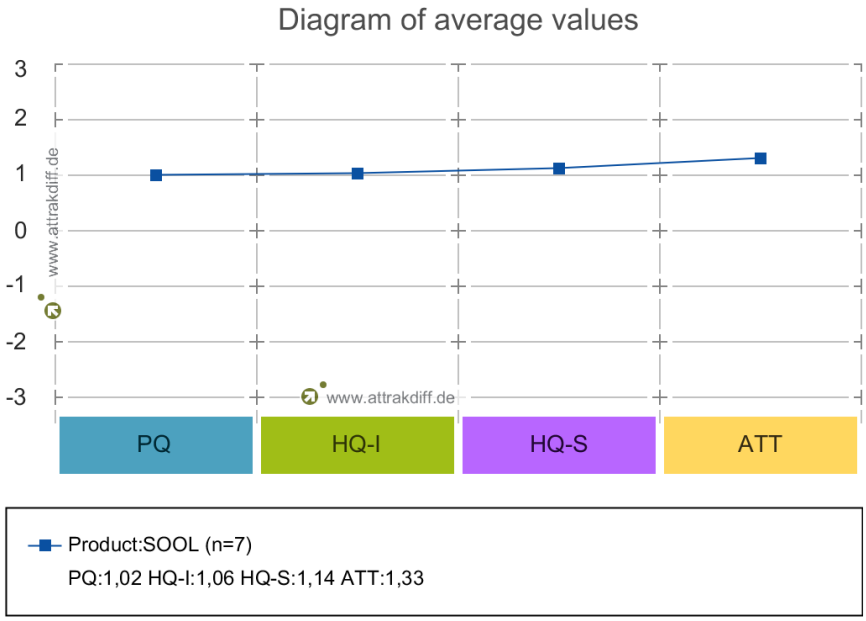


Figure 6.8 AttrakDiff results per category

The analysis of the AttrakDiff questionnaire's answers shows good ratings for the UX of SOOL. Likert scale results are categorized into negative (-3 to -1), neutral (0) and positive (1 to 3). The app scored positively in all 4 categories.

The participants rated the pragmatic qualities with an average score of 1.02 (SD=1.493), showing above average effectiveness and efficiency. While there is still room for improvements, this signifies generally good usability (see figure 6.8).

Hedonistic aspects have been rated with an average score of 1.10 (SD=1.410), with stimulation rating slightly higher (M=1.143, SD=1.291) than identity (M=1.061, SD=1.533). Similar to the pragmatic rating, these scores indicate that the application is appealing to users (see figure 6.8).

Attractiveness scored the highest of all attributes with a score of 1.33 (SD=1.297) (see figure 6.8).

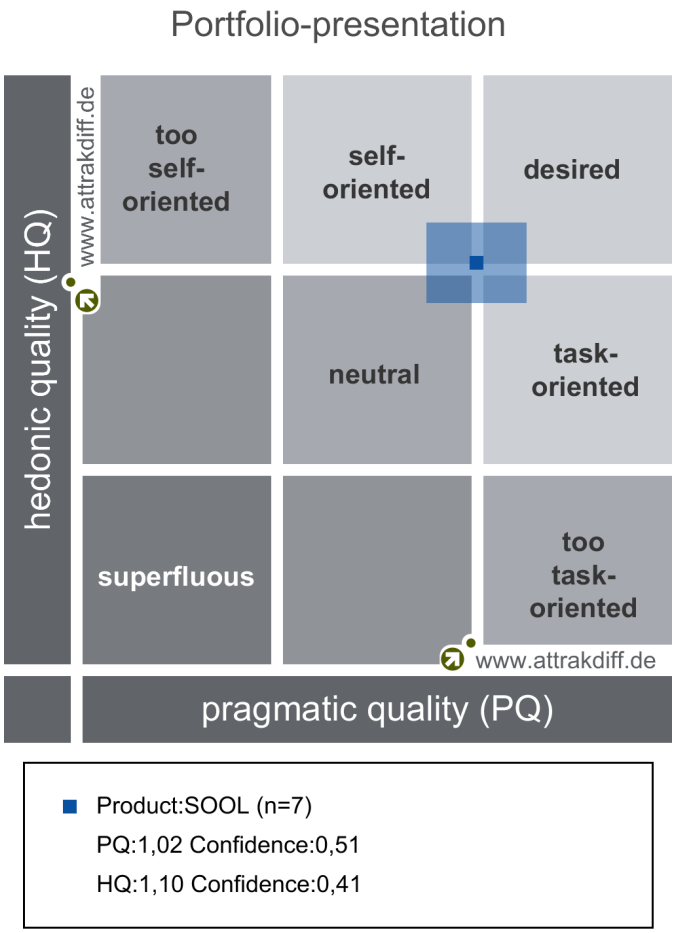


Figure 6.9 AttrakDiff results per word pair

According to Hassenzahl et al. (2008) these results show, that the overall UX of SOOL is very attractive. As seen in figure 6.9, the UX of the app is balanced between being self-oriented and task-oriented, equally bordering between neutral and desired qualities.

As seen in figure 6.10, the app gained high scores for the word pairs 'isolating-connective', 'alienating-integrating', 'conventional-inventive', 'ordinary-novel' and 'impractical-practical', suggesting success in our research goal to generate a new and rewarding user experience.. The only

negative scores were given 'unprofessional-professional', 'undemanding-challenging' and 'ugly-attractive' and can be attested to the early stage in our prototype development.

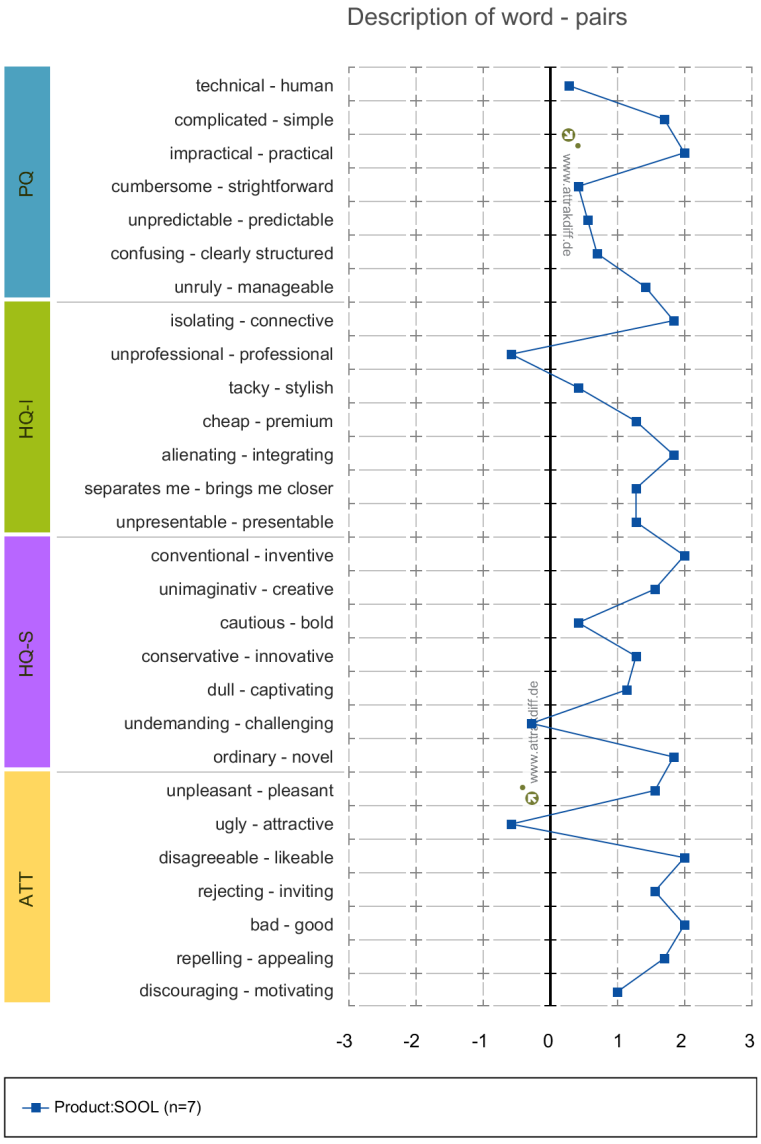


Figure 6.10 AttrakDiff results per word pair

6.4 IOS

The IOS did not show any significant differences between the felt closeness of the participants before or after the study. Before the study, the participants rated the subjective closeness to

their flatmates with an average score of 4.250 (SD=1.628), while after the study, the reported closeness was 4.6% higher with an average score of 4.444 (SD=1.611) (see figure 6.11).

Table 6.2 IOS Descriptives

	N	Mean	SD	SE
Before	36	4.250	1.628	0.271
After	36	4.444	1.611	0.269

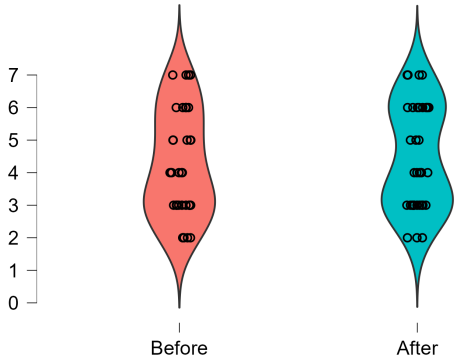


Figure 6.11 IOS Scale data plot

Table 6.3 IOS Paired Samples T-Test (all)

Measure 1		Measure 2	t	df	p	Mean Difference	SE Difference
Before	-	After	−1.557	35	0.128	−0.194	0.125

A paired t-test (see table 6.3) proved no significant differences between the two data sets (t(35)=-1.557, p= 0.128). Further analysis of the data showed that 3 participants gave identical scores before and after. One participant's data scored a p-value of 1 in a paired sample t-test. This leaves 3 participants with traceable differences, one of which proved to possess statistically significant differences between the two conditions (t(6)=-2.646, p=0.035) (see table 6.4).

Table 6.4 IOS Paired Samples T-Tests (individually)

Measure 1		Measure 2	t	df	p
Alias 1	-	Alias 1 (after)	0.000	6	1.000
Alias 2	-	Alias 2 (after)	−0.281	6	0.788
Alias 3	-	Alias 3 (after)	−2.646	6	0.038
Alias 4	-	Alias 4 (after)	−1.000	6	0.356

6.5 Interview

Interviews were conducted in German, citations have been translated to English.

One participant was not comfortable with being recorded and was therefore excluded from the interview analysis, leaving 6 interviewed participants.

The app and its underlying concepts gained high praise from the participants. They found it to be "innovative", "exciting", "cool" and "funny".

> *"I like the concept of the app very, very much. [...] I think it's very innovative in the way that I've never seen anything of the sort anywhere else and think it's very nice, the underlying idea of sitting in the bus and then to connect - without conversation or small talk - simply through the music."* (Participant 5)

When asked about their specific song choices and their motivation for sharing them, the participants all admitted to having consciously chosen tracks they deemed presentable to the group. 5 Participants reported having chosen primarily funny songs and titles that should catch the eye of other participants and entertain them.

> *"It is somehow funny when you start to consider differently what to throw in there next. When you know which other people are currently using the app, you contemplate [...] what do I want to share? What is especially funny or maybe also especially catchy so that people think: I will click on this because it has an exciting title or a funny album cover."* (Participant 7)

4 participants reported that they put on songs they personally listen to and like to show them to users that might also like them.

2 participants reported that the music shared over the app is potentially of another character than music shared in person. Anonymously shared music doesn't have to fit a specific setting or mood or necessarily be presentable. Other users are free to explore the song choices privately, devoid of social pressure.

> *"Well, you usually know the musical tastes of other people. What they normally put on. But now you had a new kind of interest because people on their own - or also in this kind of sharing context - listen to different music than you would put on if you were to listen to music together in one room, over the same speakers or the same playback device. Because that's when you would put on specific kinds of music that fit the situation where you, for example, might sit together and drink alcohol or clean up together or are just talking somehow. There is different music playing than you would put on when listening to music on your own."* (Participant 7)

One participant admitted to at some point putting on random songs from their playlist without further moderation. Another participant stated that she would use the app to share artists she wanted to support.

> *"I think if I really were sitting in the tram and it wasn't just a small study we made in the shared flat, I think I'd definitely put on what I like to listen to the most because that's what I would want to share the most. [...] [Really more my favorite songs where I would be inclined to pass them on.] The app could definitely encourage you to support artists that you like."* (Participant 5)

Most participants exchanged thoughts about the song choices present during the study.

> *"We made fun of the song titles, looked in which albums and playlists they appear and assigned the music to the individual listeners. 'So you listen to this and that.', 'Did you just put this on?', 'Someone is listening to 'Ich Hasse Music' (I Hate Music).', 'Oh that's Participant 7.'"* (Participant 7)
>
> *"Through that, I discovered that song that was named 'Die Wirkliche Männlichkeit' (The True Manliness) to which Participant 4 was listening, and I was like what? And then we started talking about it, and he said: yes, it's actually a good song that is questioning and challenging stereotypical manliness. And that would not have happened without the app."* (Participant 1)

5 participants stated that through sharing and listening, they felt more connected with the other participants and their surroundings.

> *"Well, it's a different form of connectedness than when you are talking or really getting to know someone but rather some kind of subliminal connectedness through music that is somehow very pleasant and without forced social contact." - Participant 5*

One participant stated that she did not feel more connected with the other participants but probably would if they had been strangers.

> *"I didn't really feel a difference, but if it would have been strangers, I think I would."* (Participant 6)

One participant mentioned that she felt that music always has the ability to connect, even if shared with random strangers.

> *"I actually have had a similar experience in real life, where I saw some girl at a traffic light that was dancing with her music and I saw her again later, still dancing and I spoke to her and asked her: Wow, what are you listening to? Since then, I've been listening to this song from time to time and always remember this beautiful situation. And the girl. It was a very nice vibe, but I think most people wouldn't directly address somebody - especially not randomly in the streets. But this makes the step easier and gives you good memories of the girl."* (Participant 1)

All participants stated that they would likely use the app if it was available and possessed a sufficient user base. As motivations for using the app, participants listed entertainment, bridging time, discovering new music, feeling more connected with their surroundings and connecting to other people.

> *"It's an app with which you can be on the move, you can kill time without being completely closed off from all other people you see at the moment, but instead engage in your surroundings which, especially in the city context - for example when you're using public transport - is normally not a given thing and which I don't know from any other app."* (Participant 7)
>
> *"You gain a little insight into a mobile phone that you usually don't have."* (Participant 5)

All participants were positive that SOOL could help them discover new music and some of them already did in the course of the study.

> *"I liked it, and I even memorized some stuff. I even made a playlist with things I found cool."* (Participant 6)
>
> *"You get to titles, albums and genres that either wouldn't have crossed your mind or that you can't really search for if you didn't know of them before. And that gives you new insights into all that is out there in the world of music."* (Participant 3)

3 participants mentioned that it would be nice if the Spotify playback could be integrated into SOOL so you would not have to switch apps.

> *"To keep the threshold low, it would be cool if you only had to open one app. And not Spotify and additionally the app. If you had a plugin in Spotify or in SOOL to start the respective other one. [...] It would be nice if you said, 'OK, I want to listen to music and share it' and you only had to press once. People are lazy."* (Participant 7)

Multiple participants wished for some way of interacting with other users. Some wished for feedback when someone listened to their shared songs or some sort of 'like' function. One participant wished for a chat function to connect to people that share the same taste.

> *"It would be cool if people could see that, when you like their stuff and then know: 'OK, people liked what I put on right now'. That would actually be very cool."* (Participant 5)
>
> *"A chat would be cool. [...] It's a small change that would achieve a totally different outcome. Well, one additional outcome - that you can actually connect. [...] If I'd be sitting in the tram, listening to my absolutely favorite niche song, that no one else knows or something, and someone was listening to the same, I'd definitely write that person."* (Participant 3)

The participants showed some disparity considering aesthetic design aspects of the app. Two participants liked the plain and functional style of the app, while two found it rather boring and wished for a more polished look.

> *"Otherwise, I think the interface is beautiful - especially because it's so minimalist and you don't have the feeling that you have a shiny ... I don't know ... smooth-edged app interface."* (Participant 7)
>
> *"I liked the design. It's point-blank. But I don't know if it appeals to everyone if it is so super easy. If it looks professional, having only these simple parts. Sometimes it needs some surrounding stuff to look WOW."* (Participant 1)

Participants reported that most of the elements in the app were self-explanatory. Still, they expressed doubts that, had it not been for the introduction given at the beginning of the study, they would have been able to grasp all presented functions and mechanisms the app presents. They suggested some sort of tutorial before using the app for the first time that would explain the main mechanisms, icons and menu points.

> *"I don't know if I had known how to get it running and where all particular things were without guidance. Many of the functions were somehow intuitive, but with one or two things, I don't know if I would have found them instantly and without further explanation. "* (Participant 7)

One participant wished for a reminder outside the app that you are still sharing (e.g. in the status bar or on the lock screen).

> *"I would fear that I forget that I am sharing and then listen to my embarrassing insecurity songs, and then all my flatmates could listen in."* (Participant 1)

Participants further mentioned that in a finished app, they would wish for some means of customization, for example, color themes, language and user profiles.

One participant wished for some means of localizing other users through some sort of compass to also be able to connect in person.

Chapter 7

Discussion

In this thesis, we aimed to accumulate existing concepts for location-based music listening applications to provide orientation for other researchers. In our own study, we combined found key features with modern technology to create a novel, rewarding application that has the aptitude for creating anonymous social interactions, fostering feelings of connectedness in public environments and creating a platform for music enthusiasts to discover and share music and express themselves. In the following chapter, we will discuss the findings of our user study, and its relevance for future research.

7.1 About the Results

We would like to address the validity of our findings in relation to our group of participants.

Similar to other user studies in the field, our initial user study consisted of a rather small group of 7 participants. Our participants were all recruited from the same shared apartment, where all but one student and all in a similar age group (20-28, M=24). It should be noted that, given their similar demographics, the participants don't represent all potential users, but we further like to argue that 55% of all Spotify users are in the age group between 18 and 35, placing our participants well inside the target user group (*50+ Statistics Proving Spotify Growth is Soaring in 2021* 2021).

According to Nielsen et al. (1993) a number of 5 participants can discover over 80% of all usability problems. This is sufficient for our iterative approach to designing SOOL. Additionally, we argue that our aim was to gain qualitative data from our study and focused on the idiocratic features of our users' experiences. We gathered significant insight from the participant's accounts concerning our approach to improve mobile music listening experiences in public areas.

7.2 Research Questions

For our study, we set 3 research questions with a focus on confirming the validity and aptitude of our concept.

> **Q1:** *Is the app engaging and fun to use?*

The analysis of the AttrakDiff questionnaire results shows that SOOL was positively received in terms of pragmatic and hedonic qualities as well as attractiveness. Comments gathered during the interview showed that the users perceived the use of the app as highly entertaining and interesting. They expressed wishes to use it in their daily life.

We argued that these findings show that SOOL can provide a captivating and entertaining experience for users.

> **Q2:** *Does the proposed concept provide novel forms of interaction and fosters more connectedness among users?*

Based on the interviews, the participants found the app to be innovative and mentioned that they had never encountered a similar concept. They further reported that they felt more connected to other users and saw the concepts potential to connect to strangers through the anonymous sharing of music. The IOS questionnaire showed no significant difference in felt closeness before and after the study. We argue that this is mainly due to the short duration of the study and the fact that participants already knew each other.

Based on these results, it can be argued that we succeeded in creating novel and satisfying forms of interaction. Results show that the app has a great potential to connect users.

> **Q2:** *Does the concept provide means to escape the musical filter bubble and discover new music?*

During the study, the participants vigorously shared songs. They reported that the concept of SOOL would positively enable them to discover new music and explore different genres outside their normal scope. Some reported to already have discovered new music during the study that they memorized and will listen to again. As mentioned above, participants stated to have never encountered this way of sharing.

We therefore argue that the concept fosters serendipitous music encounters and constantly exposes users to diverse musical tastes outside of their usual bubble.

7.3 Conclusion

Modern music listening technologies have transformed the way we consume music in our everyday lives. While music has become wider, faster and more easily accessible, consumption practices have shifted from public shared consumption to private consumption. We identified factors in modern mobile music consumption limiting social interactions and exchange as well as serendipitous music discovery. We presented existing research engaging these factors and analyzed the different approaches. Our findings show common concepts that we used to devise universal best practices and design paradigms for designing music listening applications focused on improving mobile listening experiences, engaging the isolating effects of mobile listening practices and encouraging social interactions in public spaces.

Our study of existing research further revealed that the field of location-based mobile music listening applications is still largely under-researched and mostly lacks the analysis and integration of modern listening practices and technologies like smartphones or music streaming services. We therefore developed a prototype application - SOOL - which incorporates a selection of concepts from our research review as well as state-of-the-art technology. We conducted a user study to validate the chosen concepts and technical realization. The results show a very positive reception for the implemented concepts. The findings from our literature review and the evaluation of our own concept application offer valuable insight into designing context-aware mobile music listening applications and provide orientation for future research in this field.

7.4 Future Work

Due to the COVID-19 pandemic and subsequent contact restrictions, we felt a user study using any amount of strangers to be inappropriate and morally questionable at the time. In our user study, we used household members, as they constitute a steady user presence at their domicile. For future research, we propose a larger scaled user study, further studying the effects on the participants' perceived connectedness as well as the impact on listening habits. Like most PNAs, to function in the wild, SOOL would need a large existing user base to provide a sufficiently high chance of serendipitous encounters with strangers when navigating through crowded public places. Seeburger et al. (2012) and Åman et al. (2015) showed that for future user studies, it might be feasible to simulate large user bases with users facilitated or represented by researchers. This could give participants a better impression of the aspired interaction dynamics in the early development and distribution phases without having to recruit huge amounts of participants.

While the implemented prototype already shows the great potential of the studied concepts, more concepts remain to be explored. Based on our review of existing research as well as the participants' comments, future iterations of the application could include direct and indirect

communication functionality with other users (e.g. a chat function, gifting of likes and displaying listener counts), asynchronous interaction (e.g. 'dropping' of songs at specific locations), searchable user histories and profiles, sharing of playlists as well as multiple other concepts that might positively add to the user experience. In the context of our design study and the scope of this thesis, however, this was unfeasible, and we put our focus on a chosen selection of concepts.

Future iterations of SOOL will incorporate findings gathered in the user study to further improve the usability and user experience of the application. Besides revising the user interface and adding more functionality, we would also like to rework some technical aspects to improve the overall efficiency and resilience. One major concern is the overhaul of our data retrieval concept. For our prototype, we decided to use short polling (client-side requesting of data in short intervals) to retrieve data from Spotify's Web API and our database. While this was feasible for first iterations and proof-of-concept, server-side pushing in reaction to changes in the database (e.g. realized with the WebSocket API or Server-Sent Events (SSEs)) enables far better accuracy concerning synchronicity and data handling.

Appendix A

Appendix

A.1 Questionnaires

A.1.1 Demographics Questionnaire

Table A.1 Demographics Questionnaire Items

ID	Question	Answer options
Q1	How old are you?	number
Q2	What is your gender?	male, female, non-binary, prefer not to say, other
Q3	What is your occupation?	Student, employed full-time, employed half-time, self-employed, retired, unemployed, other

A.1.2 Listening Preferences Questionnaire

Table A.2 Listening Preferences Questionnaire Items

ID	Question	Answer options
Q1	How many days a week do you listen to music on Spotify?	Likert scale 0-7

ID	Question	Answer options
Q2	How long do you listen to music on Spotify every day?	<1hour, 1-2hours, 3-4hours,>4hours
Q3	How do you listen to music?	Streaming from my own playlists, Streaming playlists made by others, Automatically created playlists, streaming albums, streaming songs, other
Q4	How do you discover new music?	streaming service recommendations (Spotify, YouTube, Soundcloud, Last.FM, etc.), Streaming service playlists and mixes, friend's recommendations, radio, blogs and critics, local live concerts, self promotion of bands online (Reddit, Facebook, etc.), other
Q5	How many days a week do you visit public urban environments?	Likert scale 0-7
Q6	How likely do you listen to music when alone in public environments?	Likert Scale 1-5, never→ always
Q7	Do you ever wonder, what other people are listening to in public environments?	Likert Scale 1-5, never→ always
Q8	Why do you listen to music in public environments?	distraction, entertainment, relaxation, preparation, other

A.1.3 AttrakDiff Questionnaire

Table A.3 AttrakDiff Questionnaire(Burmester et al., 2011)

ID		1	2	3	4	5	6	7	
Q1	human	☐	☐	☐	☐	☐	☐	☐	technical
Q2	isolating	☐	☐	☐	☐	☐	☐	☐	connective

ID		1	2	3	4	5	6	7	
Q3	pleasant	☐	☐	☐	☐	☐	☐	☐	unpleasant
Q4	inventive	☐	☐	☐	☐	☐	☐	☐	conventional
Q5	simple	☐	☐	☐	☐	☐	☐	☐	complicated
Q6	professional	☐	☐	☐	☐	☐	☐	☐	unprofessional
Q7	ugly	☐	☐	☐	☐	☐	☐	☐	attractive
Q8	practical	☐	☐	☐	☐	☐	☐	☐	impractical
Q9	likable	☐	☐	☐	☐	☐	☐	☐	disagreeable
Q10	cumbersome	☐	☐	☐	☐	☐	☐	☐	straightforward
Q11	stylish	☐	☐	☐	☐	☐	☐	☐	tacky
Q12	predictable	☐	☐	☐	☐	☐	☐	☐	unpredictable
Q13	cheap	☐	☐	☐	☐	☐	☐	☐	premium
Q14	alienating	☐	☐	☐	☐	☐	☐	☐	integrating
Q15	brings me closer to people	☐	☐	☐	☐	☐	☐	☐	seperates me from people
Q16	unpresentable	☐	☐	☐	☐	☐	☐	☐	presentable
Q17	rejecting	☐	☐	☐	☐	☐	☐	☐	inviting
Q18	unimaginative	☐	☐	☐	☐	☐	☐	☐	creative
Q19	good	☐	☐	☐	☐	☐	☐	☐	bad
Q20	confusing	☐	☐	☐	☐	☐	☐	☐	clearly structured
Q21	repelling	☐	☐	☐	☐	☐	☐	☐	appealing
Q22	bold	☐	☐	☐	☐	☐	☐	☐	cautious
Q23	innovative	☐	☐	☐	☐	☐	☐	☐	conservative
Q24	dull	☐	☐	☐	☐	☐	☐	☐	captivating
Q25	undemanding	☐	☐	☐	☐	☐	☐	☐	challenging
Q26	motivating	☐	☐	☐	☐	☐	☐	☐	discouraging

ID		1	2	3	4	5	6	7	
Q27	novel	☐	☐	☐	☐	☐	☐	☐	ordinary
Q28	unruly	☐	☐	☐	☐	☐	☐	☐	manageable

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A.4 List of Abbreviations

AR Augmented Reality

CD Compact Disc
CLI command-line interface

GPS Global Positioning System

HCI Human-Computer Interaction

IOS Scale Inclusion of Others in Self Scale

JSON JavaScript Object Notation

LP Long Play

MP3 MPEG (Motion Picture Experts Group) Layer-3

PDA Personal Digital Assistant

PNA People-Nearby Application

SOOL Soundtrack Of Our Lives
SSEs Server-Sent Events

UI User Interface
URL Uniform Resource Locator
UX user experience

WoZ Wizard of Oz

A.5 Glossary

API
(application programming interface) - a software interface that offers a service to other pieces of software (pp. 31, 33, 37, 58)

APK
(Android application package) - file format used by the Android operating system to install new applications (p. 26)

App Store
a digital distribution platform for iOS applications (p. 26)

Apple Music
a music and video streaming service developed by Apple (p. 7)

Expo
an open-source platform for making universal native apps for Android, iOS, and the web with JavaScript and React (pp. 26, 33)

Express.js
the standard server framework for Node.js (pp. 33, 37)

GeoJSON
an open standard format designed for representing simple geographical features, along with their non-spatial attributes (p. 35)

Git
an open source distributed version control system (p. 33)

Heroku
a cloud platform as a service supporting several programming languages (p. 33)

iPhone
a line of smartphones designed and marketed by Apple (p. 9)

iPod
a line of portable media players and multi-purpose mobile devices designed and marketed by Apple (pp. 6, 7, 9)

iTunes Store
an online digital media store operated by Apple (p. 7)

MongoDB
an open source NoSQL database management program (p. 33)

Napster
a peer-to-peer download platform (p. 7)

Node.js
an open-source and cross-platform JavaScript runtime environment (pp. 26, 33, 37)

OAuth 2.0
the industry-standard protocol for authorization (p. 37)

Pandora
a music streaming service (p. 7)

Pokémon Go
an augmented reality (AR) mobile game released in 2016 (p. 20)

REST
(REpresentational State Transfer) - a software architectural style that defines principles for the design of internet hypermedia systems (p. 37)

Spotify
the most popular music streaming service (pp. 2, 7, 9, 10, 14, 23, 25–29, 31–33, 35–37, 39–41, 43, 44, 46, 53, 55, 58, 62)

Appendix B

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